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MODERN INSTITUTE OF TECHNOLOGY & RESEARCH CENTRE, ALWAR

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Subject & Subject Code: BEE, 2FY2-07

Semester: II

Session: 2022-23 (Even Sem.)

Branch: First Year

Batch: D

Unit: I

Date of Submission: 20/03/23

Lecture No.	Topics Covered	Page No.	Total Page
1	Basic Concepts of Electric Current & Electric Power	1-4	3
2	Ohms Law & Active, Passive Sources	5-8	3
3	Types of Network Elements	9-12	4
4	Electrical Circuit Elements(R,L,C) , Voltage & Current Source	13-16	4
5	KCL & KVL and Series-Parallel Circuit	17-20	4
6	Mesh analysis	28-31	4
7	Node Analysis	32-35	4
8	Thevenins Theorem	36-39	4
9	Superposition Theorem	40-43	4

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Lecture-1
Basic Concepts of Electric current & Electric Power

1.1 INTRODUCTION

Given an electrical network, the network analysis involves various methods. The process of finding the network variables namely the voltage and currents in various parts of the circuit is known as network analysis. Before we carry out actual analysis it is very much essential to thoroughly understand the various terms associated with the network. In this chapter we shall begin with the definition and understanding in detail some of the commonly used terms. The basic laws such as Ohm's law, KCL and KVL, those can be used to analyse a given network Analysis becomes easier if we can simplify the given network. We will be discussing various techniques, which involve combining series and parallel connections of R, L and C elements.

1.2 SYSTEMS OF UNITS

As engineers, we deal with measurable quantities. Our measurement must be communicated in standard language that virtually all professionals can understand irrespective of the country. Such an international measurement language is the International System of Units (SI). In this system, there are six principal units from which the units of all other physical quantities can be derived.

Quantity	Basic Unit	Symbol
Length	Meter	M
Mass	kilogram	kg
Time	second	s
Electric Current	ampere	A
Temperature	Kelvin	K
Luminous intensity	candela	Cd

One great advantage of SI unit is that it uses prefixes based on the power of 10 to relate larger and smaller units to the basic unit.

Multiplier	Prefix	Symbol
10^{12}	Tera	T
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	K
10^{-3}	milli	m
10^{-6}	micro	µ
10^{-9}	nano	n
10^{-12}	pico	p

1.3 BASIC CONCEPTS AND DEFINITIONS

1.3.1 CHARGE

The most basic quantity in an electric circuit is the electric charge. We all experience the effect of electric charge when we try to remove our wool sweater and have it stick to our body or walk across a carpet and receive a shock.

Charge is an electrical property of the atomic particles of which matter consists, measured in coulombs (C). Charge, positive or negative, is denoted by the letter q or Q.

We know from elementary physics that all matter is made of fundamental building blocks known as atoms and that each atom consists of electrons, protons, and neutrons. We also know that the charge 'e' on an electron is negative and equal in magnitude to 1.602×10^{-19} C, while a proton carries a positive charge of the same magnitude as the electron and the neutron has no charge. The presence of equal numbers of protons and electrons leaves an atom neutrally charged.

1.3.2 CURRENT

Current can be defined as the motion of charge through a conducting material, measured in Ampere (A). Electric current, is denoted by the letter i or I.

The unit of current is the ampere abbreviated as (A) and corresponds to the quantity of total charge that passes through an arbitrary cross section of a conducting material per unit second.
Mathematically,

$$I = \frac{Q}{t} \text{or} Q = It$$

Where Q is the symbol of charge measured in Coulombs (C), I is the current in amperes (A) and t is the time in second (s).

The current can also be defined as the rate of charge passing through a point in an electric circuit**.**
Mathematically,

$$i = \frac{dq}{dt}$$

The charge transferred between time t_1 and t_2 is obtained as

$$q = \int_{t_1}^{t_2} i dt$$

A constant current (also known as a direct current or DC) is denoted by symbol I whereas a time-varying current (also known as alternating current or AC) is represented by the symbol i or i (t). Figure 1.1 shows direct current and alternating current.

Current is always measured through a circuit element as shown in Fig. 1.1

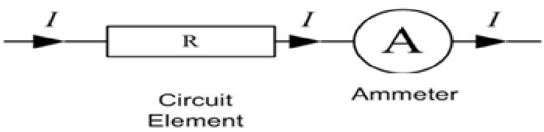


Fig. 1.1 Current through Resistor (R)

Two types of currents:
1) A direct current (DC) is a current that remains constant with time.
2) An alternating current (AC) is a current that varies with time.

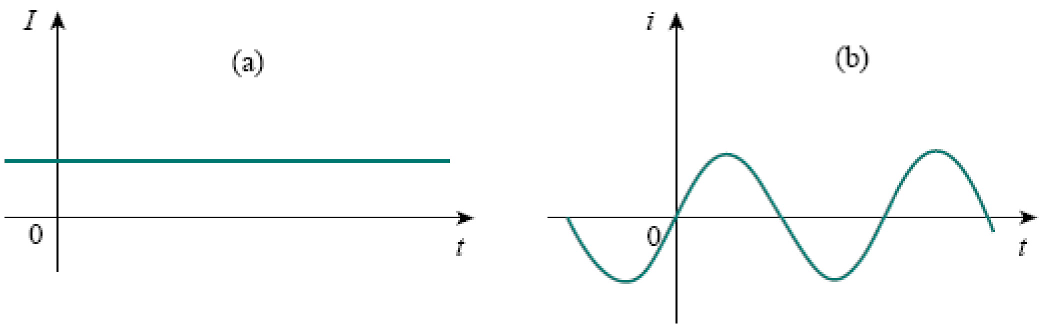


Fig. 1.2Two common types of current: (a) direct current (DC), (b) alternative current (AC)

Example 1.1
Determine the current in a circuit if a charge of 80 coulombs passes a given point in 20 seconds (s).

Solution:

$$I = \frac{Q}{t} = \frac{80}{20} = 4\ A$$

Example 1.2
How much charge is represented by 4,600 electrons?

Solution:
Each electron has - $1.602\text{x}10^{-19}$ C. Hence 4,600 electrons will have:

$-1.602\text{x}10^{-19}\text{x}4600 = -7.369\text{x}10^{-16}$ C

Example 1.3
The total charge entering a terminal is given by q=5$t$sin4πtmC. Calculate the current at t=0.5 s.

Solution:

$$i = \frac{dq}{dt} = \frac{d}{dt}(5tsin4\pi t) = (5sin4\pi t + 20\pi tcos4\pi t)\ mA$$

At t=0.5 s.
i = 31.42 mA

Example 1.4
Determine the total charge entering a terminal between t=1 s and t=2 s if the current passing the terminal is i= $(3t^2-t)$ A.
Solution:

$$q = \int_{t=1}^{t=2} idt = \int_{1}^{2}(3t^2 - t)dt = \left(t^3 - \frac{t^2}{2}\right)_1^2 = (8-2) - \left(1 - \frac{1}{2}\right) = 5.5C$$

1.3.3 VOLTAGE (or) POTENTIAL DIFFERENCE

To move the electron in a conductor in a particular direction requires some work or energy transfer. This work is performed by an external electromotive force (emf), typically represented by the battery in Fig. 1.3. This emf is also known as voltage or potential difference. The voltagev_{ab}between two points aand b in an electric circuit is the energy (or work) needed to move a unit charge from a to b.

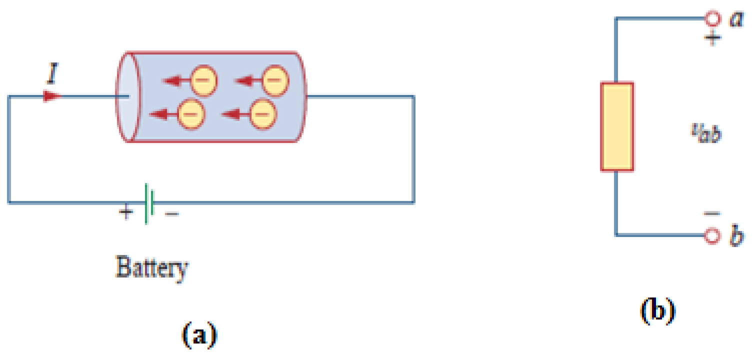


Fig. 1.3(a) Electric Current in a conductor, (b)Polarity of voltage v_{ab}

Voltage (or potential difference) is the energy required to move charge from one point to the other, measured in volts (V). Voltage is denoted by the letter v or V.
Mathematically,

$$v_{ab} = \frac{dw}{dt}$$

where w is energy in joules (J) and q is charge in coulombs (C). The voltage v_{ab} or simply V is measured in volts (V).

1 volt = 1 joule/coulomb = 1 newton-meter/coulomb

Fig. 1.3 shows the voltage across an element (represented by a rectangular block) connected to points a and b. The plus (+) and minus (-) signs are used to define reference direction or voltage polarity. The v_{ab} can be interpreted in two ways: (1) point a is at a potential of v_{ab} volts higher than point b, or (2) the potential at point a with respect to point b is v_{ab} . It follows logically that in general

$$v_{ab} = -v_{ba}$$

Voltage is always measured across a circuit element as shown in Fig. 1.4

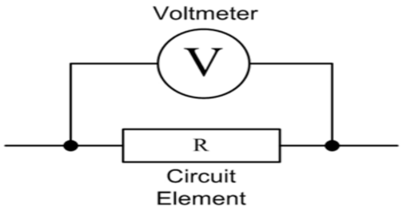


Fig. 1.4 Voltage across Resistor (R)

Example 1.5

An energy source forces a constant current of 2 A for 10 s to flow through a lightbulb. If 2.3 kJ is given off in the form of light and heat energy, calculate the voltage drop across the bulb.

Solution:

Total charge dq= i*dt = 2*10 = 20 C

The voltage drop is $v = \frac{dW}{dq} = \frac{2.3*10^3}{20} = 115\ V$

1.3. 4 POWER

Power is the time rate of expending or absorbing energy, measured in watts (W). Power, is denoted by the letter p or P.

Mathematically,

$$p = \frac{dw}{dt}$$

Where p is power in watts (W), w is energy in joules (J), and t is time in seconds (s).

From voltage and current equations, it follows that;

$$p = \frac{dw}{dt} = \frac{dw}{dq} * \frac{dq}{dt} = V * I$$

Thus, ifthe magnitude of current I and voltage are given, then power can be evaluated as the product of the two quantities and is measured in watts (W).

Sign of power:

Plus sign: Power is absorbed by the element. (Resistor, Inductor)

Minus sign: Power is supplied by the element. (Battery, Generator)

Passive sign convention:

If the current enters through the positive polarity of the voltage, p = +vi
If the current enters through the negative polarity of the voltage, p = – vi

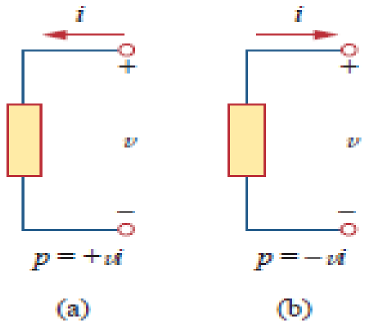


Fig 1.5 Polarities for Power using passive sign convention
(a) Absorbing Power (b) Supplying Power

1.3.5 ENERGY

Energy is the capacity to do work, and is measured in joules (J).
The energy absorbed or supplied by an element from time 0 to t is given by,

$$w = \int_0^t p dt = \int_0^t vi dt$$

The electric power utility companies measure energy in watt-hours (WH) or Kilo watt-hours (KWH)

1 WH = 3600 J

Example 1.6
A source e.m.f. of 5 V supplies a current of 3A for 10 minutes. How much energy is provided in this time?
Solution:

$$W = VIt = 5 \times 3 \times 10 \times 60 = 9\ kJ$$

Example 1.7
An electric heater consumes 1.8Mj when connected to a 250 V supply for 30 minutes. Find the power rating of the heater and the current taken from the supply.
Solution:
$P = W/t = (1.8\times10^6)/\ (30\times60) = 1000\ W$
Power rating of heater = 1kW
$P=VI$
Thus
$I=P/V=1000/250=4A$
Hence the current taken from the supply is 4A.

Example 1.7
Find the power delivered to an element at t=3 ms if the current entering its positive terminals is i=5cos60πtA and the voltage is: (a) v=3i, (b) v=3didt.
Solution:
(a) The voltage is v=3i=15cos60πtA; hence, the power is: p=vi=75cos260πtW
At t=3 ms,
p=75cos260πt×3×10^{-3} =53.48 W

(b) We find the voltage and the power as
v=3$didt$=3 −60π 5sin60πt=−900πsin60πtV
p=vi=−4500πsin60πtcos60πtW

Lecture-2
Ohms Law & Circuit Elements

1.4 OHM'S LAW

Georg Simon Ohm (1787–1854), a German physicist, is credited with finding the relationship between current and voltage for a resistor. This relationship is known as Ohm's law.

Ohm's law states that at constant temperature, the voltage (V) across a conducting material is directly proportional to the current (I) flowing through the material.
Mathematically,

$$V \alpha I$$

$$V=RI$$

Wherethe constant of proportionality R is called the resistance of the material. The V-I relation for resistor according to Ohm's law is depicted in Fig.1.6

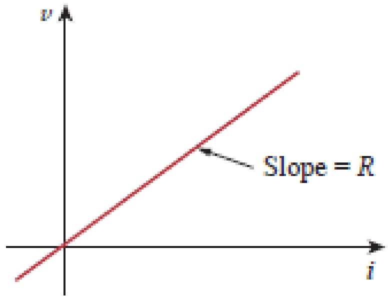


Fig. 1.6 V-I Characteristics for resistor

Limitations of Ohm's Law:

1. Ohm's law is not applicable to non-linear elements like diode, transistor etc.
2. Ohm's law is not applicable for non-metallic conductors like silicon carbide.

1.5 CIRCUIT ELEMENTS

An element is the basic building block of a circuit. An electric circuit is simply an interconnection of the elements. Circuit analysis is the process of determining voltages across (or the currents through) the elements of the circuit.

There are 2 types of elements found in electrical circuits.

a) Active elements (Energy sources): The elements which are capable of generating or delivering the energy are called active elements.
E.g., Generators, Batteries

b) Passive element (Loads): The elements which are capable of receiving the energy are called passive elements.
E.g., Resistors, Capacitors and Inductors

1.5.1 ACTIVE ELEMENTS (ENERGY SOURCES)

The energy sources which are having the capacity of generating the energy are called active elements. The most important active elements are voltage or current sources that generally deliver power/energy to the circuit connected to them.

There are two kinds of sources

a) Independent sources
b) Dependent sources

1.5.1.1 INDEPENDENT SOURCES:

An ideal independent source is an active element that provides a specified voltage or current that is completely independent of other circuit elements.

Ideal Independent Voltage Source:

An ideal independent voltage source is an active element that gives a constant voltage across its terminals irrespective of the current drawn through its terminals. In other words, an ideal independent voltage source delivers to the circuit whatever current is necessary to maintain its terminal voltage. The symbol of idea independent voltage source and its V-I characteristics are shown in Fig. 1.7

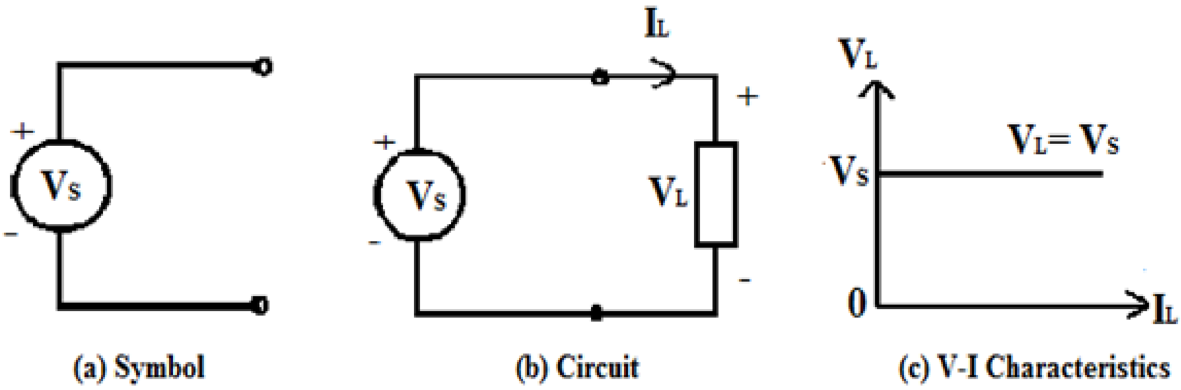


Fig. 1.7 Ideal Independent Voltage Source

Practical Independent Voltage Source:

Practically, every voltage source has some series resistance across its terminals known as internal resistance, and is represented by Rse. For ideal voltage source Rse = 0. But in practical voltage source Rse is not zero but may have small value. Because of this Rse voltage across the terminals decreases with increase in current as shown in Fig. 1.8

Terminal voltage of practical voltage source is given by

$$V_L = V_S - I_L R_{se}$$

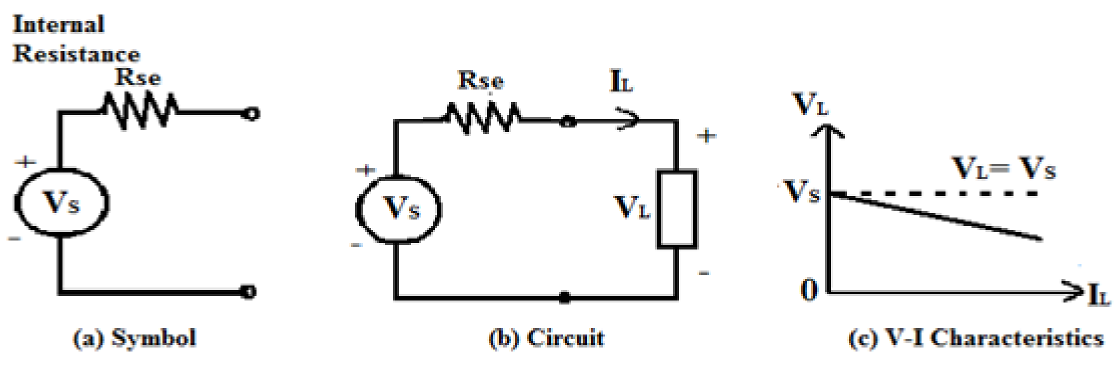


Fig. 1.8 Practical Independent Voltage Source

Ideal Independent Current Source:

An ideal independent Current source is an active element that gives a constant current through its terminals irrespective of the voltage appearing across its terminals. That is, the current source delivers to the circuit whatever voltage is necessary to maintain the designated current. The symbol of idea independent current source and its V-I characteristics are shown in Fig. 1.9

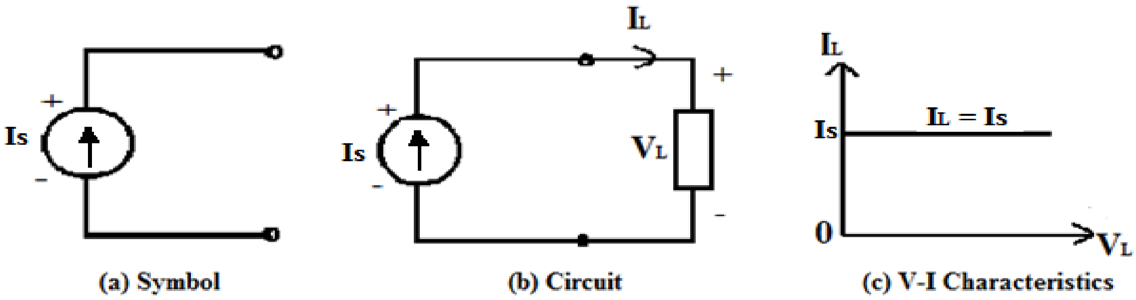


Fig. 1.9 Ideal Independent Current Source

Practical Independent Current Source:
Practically, every current source has some parallel/shunt resistance across its terminals known as internal resistance, and is represented by Rsh. For ideal current source Rsh = ∞ (infinity). But in practical voltage source Rsh is not infinity but may have a large value. Because of this Rsh current through the terminals slightly decreases with increase in voltage across its terminals as shown in Fig. 1.10.

Terminal current of practical current source is given by

$$I_L = Is - Ish$$

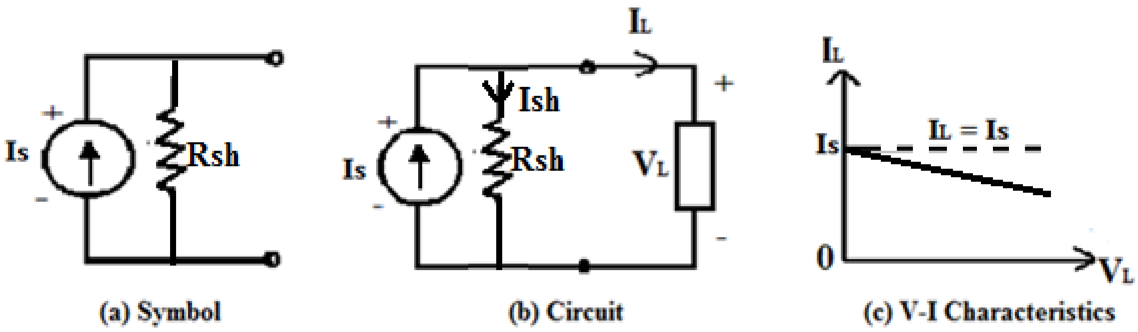


Fig. 1.10 Practical Independent Current Source

1.5.1.2 DEPENDENT (CONTROLLED) SOURCES

An ideal dependent (or controlled) source is an active element in which the source quantity is controlled by another voltage or current.

Dependent sources are usually designated by diamond-shaped symbols, as shown in Fig. 1.11. Since the control of the dependent source is achieved by a voltage or current of some other element in the circuit, and the source can be voltage or current, it follows that there are four possible types of dependent sources, namely:

1. A voltage-controlled voltage source (VCVS)
2. A current-controlled voltage source (CCVS)
3. A voltage-controlled current source (VCCS)
4. A current-controlled current source (CCCS)

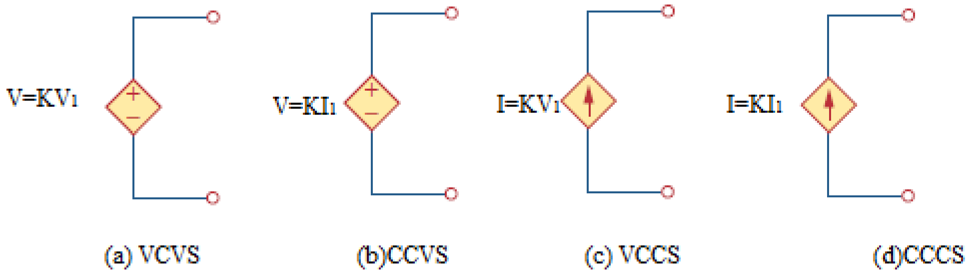


Fig. 1.11 Symbols for Dependent voltage source and Dependent current source

Dependent sources are useful in modeling elements such as transistors, operational amplifiers, and integrated circuits. An example of a current-controlled voltage source is shown on the right-hand side of Fig. 1.12, where the voltage 10i of the voltage source depends on the current i through element C. Students might be surprised that the value of the dependent voltage source is 10i V (and not 10i A) because it is a voltage source. The key idea to keep in mind is that a voltage source comes with polarities (+ -) in its symbol, while a current source comes with an arrow, irrespective of what it depends on.

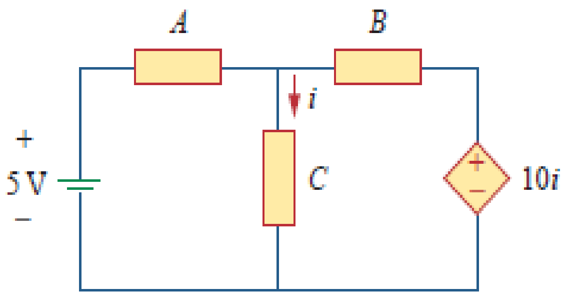


Fig. 1.12 The source in right hand side is current-controlled voltage source

1.5.2 PASSIVE ELEMENTS (LOADS)

Passive elements are those elements which are capable of receiving the energy. Some passive elements like inductors and capacitors are capable of storing a finite amount of energy, and return it later to an external element. More specifically, a passive element is defined as one that cannot supply average power that is greater than zero over an infinite time interval. Resistors, capacitors, Inductors fall in this category.

1.5.2.1 RESISTOR

Materials in general have a characteristic behavior of resisting the flow of electric charge. This physical property, or ability to resist the flow of current, is known as resistance and is represented by the symbol R.The Resistance is measured in ohms (**Ω**). The circuit element used to model the current-resisting behavior of a material is called the resistor.

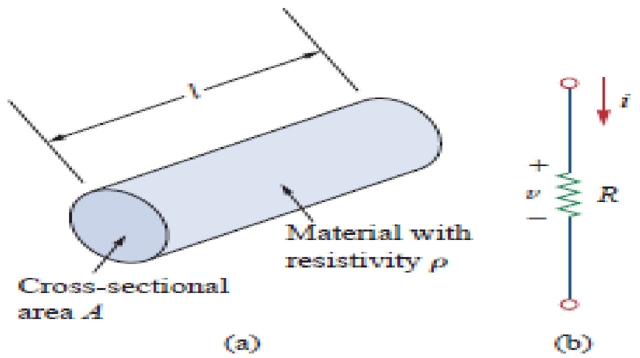


Fig. 1.13 (a) Typical Resistor, (b) Circuit Symbol for Resistor

The resistance of a resistor depends on the material of which the conductor is made and geometrical shape of the conductor. The resistance of a conductor is proportional to the its length (l) and inversely proportional to its cross sectional area (A). Therefore the resistance of a conductor can be written as,

$$R = \frac{\rho l}{A}$$

The proportionality constant ρ is called the specific resistance o resistivity of the conductor and its value depends on the material of which the conductor is made.

The inverse of the resistance is called the conductance and inverse of resistivity is called specific conductance or conductivity. The symbol used to represent the conductance is G and conductivity isσ. Thus conductivity $\sigma = 1/\rho$ and its units are Siemens per meter

$$G = \frac{1}{R} = \frac{A}{\rho l} = \frac{1}{\rho}.\frac{A}{l} = \sigma.\frac{A}{l}$$

By using Ohm's Law, The power dissipated in a resistor can be expressed in terms of R as below

$$P = VI = I^2R = \frac{V^2}{R}$$

The power dissipated by a resistor may also be expressed in terms of G as

$$P = VI = V^2G = \frac{I^2}{G}$$

The energy lost in the resistor from time 0 to t is expressed as

$$W = \int_0^t Pdt$$

$$W = \int_0^t I^2Rdt = I^2Rt$$

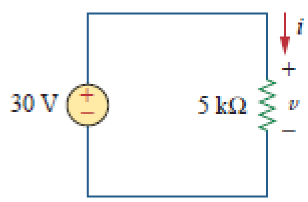
Or

$$W = \int_0^t \frac{V^2}{R}dt = \frac{V^2}{R}t$$

Where V is in volts, I is in amperes, R is in ohms, and energy W is in joules

Example 1.8

In the circuit shown in Fig. below, calculate the current i, the conductance G, the power p and energy lost in the resistor W in 2hours.



Solution:
The voltage across the resistor is the same as the source voltage (30 V) because the resistor and the voltage source are connected to the same pair of terminals. Hence, the current is

$$i = \frac{v}{R} = \frac{30}{5\text{x}10^3} = 6\,mA$$

The conductance is

$$G = \frac{1}{R} = \frac{1}{5\text{x}10^3} = 0.2mS$$

We can calculate the power in various ways

$$p = vi = 30(6\text{x}10^{-3}) = 180mW$$

or

$$p = i^2R = (6\text{x}10^{-3})^2(5\text{x}10^3) = 180mW$$

or

$$p = \frac{v^2}{R} = \frac{30^2}{5\text{x}10^3} = 180mW$$

Energy lost in the resistor is

$$W = i^2Rt = (6\text{x}10^{-3})^2(5\text{x}10^3)(2) = 360\,mWhor = 360\,mJ$$

1.5.2.2 INDUCTOR

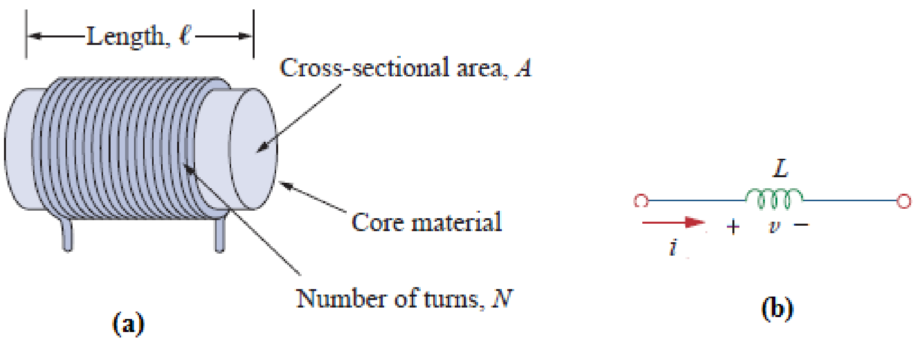


Fig. 1.14 (a) Typical Inductor, (b) Circuit symbol of Inductor

A wire of certain length, when twisted into a coil becomes a basic inductor. The symbol for inductor is shown in Fig.1.14 (b). If current is made to pass through an inductor, an electromagnetic field is formed. A change in the magnitude of the current changes the electromagnetic field. Increase in current expands the fields, and decrease in current reduces it. Therefore, a change in current produces change in the electromagnetic field, which induces a voltage across the coil according to Faraday's law of electromagnetic induction. i.e., the voltage across the inductor is directly proportional to the time rate of change of current.
Mathematically,

$$V \propto \frac{di}{dt}$$

$$v = L\frac{di}{dt}$$

Where L is the constant of proportionality called the inductance of an inductor. The unit of inductance is Henry (H).we can rewrite the above equation as

$$di = \frac{1}{L}vdt$$

Integrating both sides from time 0 to t, we get

$$\int_0^t di = \frac{1}{L}\int_0^t v\,dt$$

$$i(t) - i(0) = \frac{1}{L}\int_0^t v\,dt$$
$$i(t) = \frac{1}{L}\int_0^t v\,dt + i(0)$$
From the above equation we note that the current in an inductor is dependent upon the integral of the voltage across its terminal and the initial current in the coil$i(0)$.
The power absorbed by the inductor is
$$P = vi = Li\frac{di}{dt}$$
The energy stored by the inductor is

$$W = \int_0^t Pdt$$
$$= \int_0^t Li\frac{di}{dt}dt = \frac{Li^2}{2}$$

From the above discussion, we can conclude the following.

1. The induced voltage across an inductor is zero if the current through it is constant. That means an inductor acts as short circuit to DC.
2. A small change in current within zero time through an inductor gives an infinite voltage across the inductor, which is physically impossible. In a fixed inductor the current cannot change abruptly i.e., the inductor opposes the sudden changes in currents.
3. The inductor can store finite amount of energy. Even if the voltage across the inductor is zero
4. A pure inductor never dissipates energy, only stores it. That is why it is also called a non-dissipative passive element. However, physical inductors dissipate power due to internal resistance.

Example 1.9
Find the current through a 5-H inductor if the voltage across it is
$$v(t) = \begin{cases} 30t^2, & t > 0 \\ 0, & t < 0 \end{cases}$$
Also, find the energy stored at t = 5 s. assume initial conditions to be zero.

Solution:
$$i(t) = \frac{1}{L}\int_0^t v\,dt + i(0) = i(t) = \frac{1}{5}\int_0^t 30t^2\,dt + 0 = 6\text{x}\frac{t^3}{3} = 2t^3$$
The power $P = vi = 60t^5$

Then the energy stored is
$$W = \int_0^t Pdt = \int_0^5 60t^5 dt = 60\text{x}\left.\frac{t^6}{6}\right|_0^5 = 156.25\ kJ$$

1.5.2.2 CAPACITOR

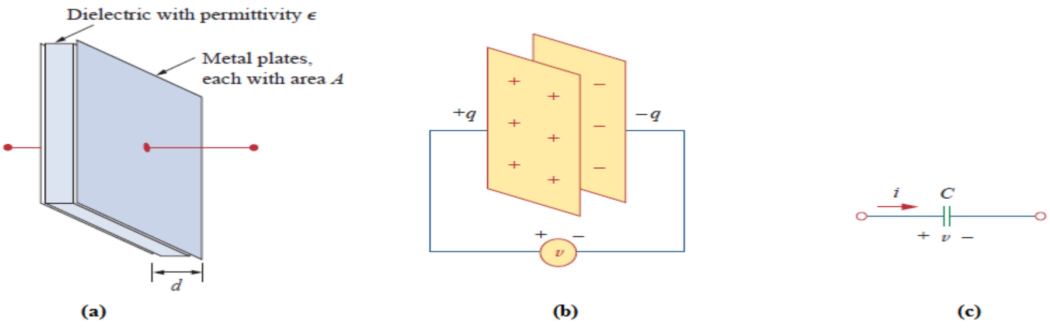


Fig. 1.15 (a) Typical Capacitor, (b) Capacitor connected to a voltage source, (c) Circuit Symbol of capacitor

Any two conducting surfaces separated by an insulating medium exhibit the property of a capacitor. The conducting surfaces are called electrodes, and the insulating medium is called dielectric. A capacitor stores energy in the form of an electric field that is established by the opposite charges on the two electrodes. The electric field is represented by lines of force between the positive and negative charges, and is concentrated within the dielectric.

When a voltage source v is connected to the capacitor, as in Fig 1.15 (c), the source deposits a positive charge q on one plate and a negative charge — q on the other. The capacitor is said to store the electric charge. The amount of charge stored, represented by q, is directly pro-proportional to the applied voltage v so that

$$q = Cv$$

Where C, the constant of proportionality, is known as the capacitance of the capacitor. The unit of capacitance is the farad (F).

Although the capacitance C of a capacitor is the ratio of the charge q per plate to the applied voltage v, it does not depend on q or v. It depends on the physical dimensions of the capacitor. For example, for the parallel-plate capacitor shown in Fig.1.15 (a), the capacitance is given by

$$C = \frac{\epsilon A}{d}$$

Where A is the surface area of each plate, d is the distance between the plates, and ϵ is the permittivity of the dielectric material between the plates.
The current flowing through the capacitor is given by

$$i = \frac{dq}{dt}$$

$$i = C\frac{dv}{dt}$$

We can rewrite the above equation as

$$dv = \frac{1}{C} idt$$

Integrating both sides from time 0 to t, we get

$$\int_0^t dv = \frac{1}{C}\int_0^t i\,dt$$

$$v(t) - v(0) = \frac{1}{C}\int_0^t i\,dt$$

$$v(t) = \frac{1}{C}\int_0^t i\,dt + v(0)$$

From the above equation we note that the voltage across the terminals of a capacitor is dependent upon the integral of the current through it and the initial voltage$v(0)$.
The power absorbed by the capacitor is

$$P = vi = vC\frac{dv}{dt}$$

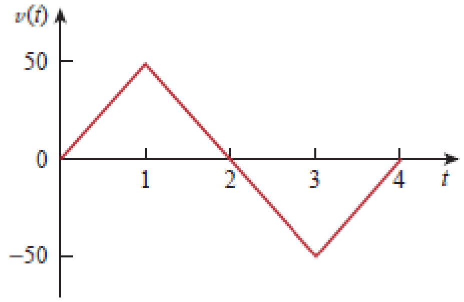
The energy stored by the capacitor is

$$W = \int_0^t Pdt$$
$$= \int_0^t vC\frac{dv}{dt}dt = \frac{Cv^2}{2}$$

From the above discussion we can conclude the following,

1. The current in a capacitor is zero if the voltage across it is constant; that means, the capacitor acts as an open circuit to DC.
2. A small change in voltage across a capacitance within zero time gives an infinite current through the capacitor, which is physically impossible. In a fixed capacitance the voltage cannot change abruptly. i.e., A capacitor will oppose the sudden changes in voltages.
3. The capacitor can store a finite amount of energy, even if the current through it is zero.
4. A pure capacitor never dissipates energy, but only stores it; that is why it is called non-dissipative passive element. However, physical capacitors dissipate power due to internal resistance.

Example 1.10
Determine the current through a $200\mu F$ capacitor whose voltage is shown in Fig. below



Solution:
The voltage waveform can be described mathematically as

$$v(t) = \begin{cases} 50tV & 0 < t < 1 \\ 100 - 50tV & 1 < t < 3 \\ -200 + 50tV & 3 < t < 4 \\ 0 & otherwise \end{cases}$$

Since $i = C\frac{dv}{dt} and C = 200\mu F,$ we take the derivative of $v(t)$ to obtain the i(t)

$$i(t) = 200\text{x}10^{-6}\text{x}\begin{cases} 50 & 0 < t < 1 \\ -50 & 1 < t < 3 \\ 50 & 3 < t < 4 \\ 0 & otherwise \end{cases}$$
$$= \begin{cases} 10\ mA & 0 < t < 1 \\ -10\ mA & 1 < t < 3 \\ 10\ mA & 3 < t < 4 \\ 0 & otherwise \end{cases}$$

Hence, the current wave form is as shown in the fig. below

Types of Network Elements

Active Elements and Passive Elements

We can classify the Network elements into either active or passive based on the ability of delivering power.

- Active Elements deliver power to other elements, which are present in an electric circuit. Sometimes, they may absorb the power like passive elements. That means active elements have the capability of both delivering and absorbing power.
  Examples: Voltage sources and current sources.
- Passive Elements can't deliver power (energy) to other elements, however they can absorb power. That means these elements either dissipate power in the form of heat or store energy in the form of either magnetic field or electric field.
  Examples: Resistors, Inductors, and capacitors.

Linear Elements and Non-Linear Elements

We can classify the network elements as linear or non-linear based on their characteristic to obey the property of linearity.

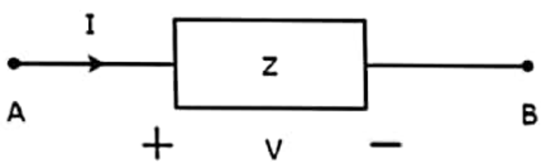
- Linear Elements are the elements that show a linear relationship between voltage and current. Examples: Resistors, Inductors, and capacitors.
- Non-Linear Elements are those that do not show a linear relation between voltage and current. Examples: Voltage sources and current sources.

Bilateral Elements and Unilateral Elements

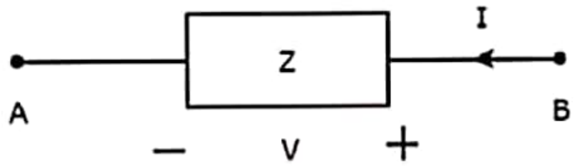
Network elements can also be classified as either bilateral or unilateral based on the direction of current flows through the network elements.

Bilateral Elements are the elements that allow the current in both directions and offer the same impedance in either direction of current flow. Examples: Resistors, Inductors and capacitors.

The concept of Bilateral elements is illustrated in the following figures.



In the above figure, the current (I) is flowing from terminals A to B through a passive element having impedance of Z Ω. It is the ratio of voltage (V) across that element between terminals A & B and current (I).



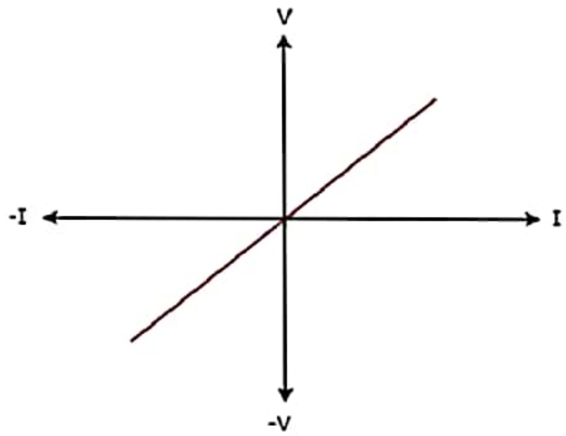
In the above figure, the current (I) is flowing from terminals B to A through a passive element having impedance of Z Ω. That means the current (–I) is flowing from terminals A to B. In this case too, we will get the same impedance value, since both the current and voltage having negative signs with respect to terminals A & B.

Unilateral Elements are those that allow the current in only one direction. Hence, they offer different impedances in both directions.

We discussed the types of network elements in the previous chapter. Now, let us identify the nature of network elements from the V-I characteristics given in the following examples.

Example 1

The V-I characteristics of a network element is shown below.



Step 1 – Verifying the network element as linear or non-linear.

From the above figure, the V-I characteristics of a network element is a straight line passing through the origin. Hence, it is linear element.

Step 2 – Verifying the network element as active or passive.

The given V-I characteristics of a network element lies in the first and third quadrants.

- In the first quadrant, the values of both voltage (V) and current (I) are positive. So, the ratios of voltage (V) and current (I) gives positive impedance values.

- Similarly, in the third quadrant, the values of both voltage (V) and current (I) have negative values. So, the ratios of voltage (V) and current (I) produce positive impedance values.

Since, the given V-I characteristics offer positive impedance values, the network element is a Passive element.

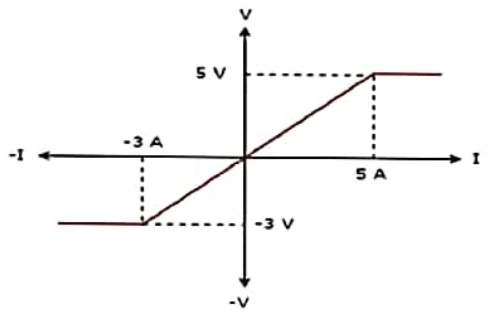
Step 3 – Verifying the network element as bilateral or unilateral.

For every point (I, V) on the characteristics, there exists a corresponding point (-I, -V) on the given characteristics. Hence, the network element is a Bilateral element.

Therefore, the given V-I characteristics show that the network element is a Linear, Passive, and Bilateral element.

Example 2

The V-I characteristics of a network element is shown below.



Step 1 – Verifying the network element as linear or non-linear.

From the above figure, the V-I characteristics of a network element is a straight line only between the points (-3A, -3V) and (5A, 5V). Beyond these points, the V-I characteristics are not following the linear relation. Hence, it is a Non-linear element.

Step 2 – Verifying the network element as active or passive.

The given V-I characteristics of a network element lies in the first and third quadrants. In these two quadrants, the ratios of voltage (V) and current (I) produce positive impedance values. Hence, the network element is a Passive element.

Step 3 – Verifying the network element as bilateral or unilateral.

Consider the point (5A, 5V) on the characteristics. The corresponding point (-5A, -3V) exists on the given characteristics instead of (-5A, -5V). Hence, the network element is a Unilateral element.

Therefore, the given V-I characteristics show that the network element is a Non-linear, Passive, and Unilateral element. The circuits containing them are called unilateral circuits.

Lumped and Distributed Elements

Lumped elements are those elements which are very small in size & in which simultaneous actions takes place. Typical lumped elements are capacitors, resistors, inductors.

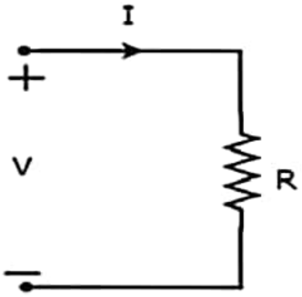
Distributed elements are those which are not electrically separable for analytical purposes.

For example a transmission line has distributed parameters along its length and may extend for hundreds of miles.

R-L-C Parameters

Resistor

The main functionality of Resistor is either opposes or restricts the flow of electric current. Hence, the resistors are used in order to limit the amount of current flow and / or dividing (sharing) voltage. Let the current flowing through the resistor is I amperes and the voltage across it is V volts. The symbol of resistor along with current, I and voltage, V are shown in the following figure.



According to Ohm's law, the voltage across resistor is the product of current flowing through it and the resistance of that resistor. Mathematically, it can be represented as

$$V = IR \qquad \text{Equation 1}$$

$$\Rightarrow I = \frac{V}{R} \qquad \text{Equation 2}$$

Where, **R** is the resistance of a resistor.

From Equation 2, we can conclude that the current flowing through the resistor is directly proportional to the applied voltage across resistor and inversely proportional to the resistance of resistor.

Power in an electric circuit element can be represented as

$$P = VI \qquad \text{Equation 3}$$

Substitute, Equation 1 in Equation 3.

$$P = (IR)I$$

$$\Rightarrow P = I^2R \qquad \text{Equation 4}$$

Substitute, Equation 2 in Equation 3.

$$P = V\left(\frac{V}{R}\right)$$

$$\Rightarrow P = \frac{V^2}{R} \qquad \text{Equation 5}$$

So, we can calculate the amount of power dissipated in the resistor by using one of the formulae mentioned in Equations 3 to 5.

Inductor

In general, inductors will have number of turns. Hence, they produce magnetic flux when current flows through it. So, the amount of total magnetic flux produced by an inductor depends on the current, I flowing through it and they have linear relationship.

Mathematically, it can be written as

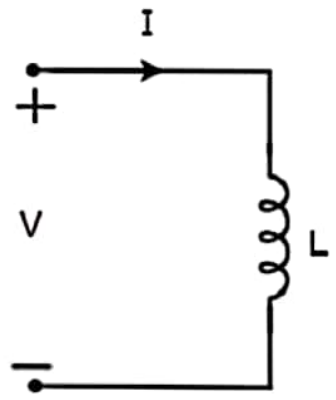
$$\Psi \,\alpha\, I$$

$$\Rightarrow \Psi = LI$$

Where,

- Ψ is the total magnetic flux
- L is the inductance of an inductor

Let the current flowing through the inductor is I amperes and the voltage across it is V volts. The symbol of inductor along with current I and voltage V are shown in the following figure.



According to Faraday's law, the voltage across the inductor can be written as

$$V = \frac{d\Psi}{dt}$$

Substitute $\Psi = LI$ in the above equation.

$$V = \frac{d(LI)}{dt}$$

$$\Rightarrow V = L\frac{dI}{dt}$$

$$\Rightarrow I = \frac{1}{L}\int Vdt$$

From the above equations, we can conclude that there exists a linear relationship between voltage across inductor and current flowing through it.

We know that power in an electric circuit element can be represented as

$$P = VI$$

Substitute $V = L\frac{dI}{dt}$ in the above equation.

$$P = (L\frac{dI}{dt})I$$

$$\Rightarrow P = LI\frac{dI}{dt}$$

By integrating the above equation, we will get the energy stored in an inductor as

$$W = \frac{1}{2}LI^2$$

So, the inductor stores the energy in the form of magnetic field.

Capacitor

In general, a capacitor has two conducting plates, separated by a dielectric medium. If positive voltage is applied across the capacitor, then it stores positive charge. Similarly, if negative voltage is applied across the capacitor, then it stores negative charge.

So, the amount of charge stored in the capacitor depends on the applied voltage **V** across it and they have linear relationship. Mathematically, it can be written as

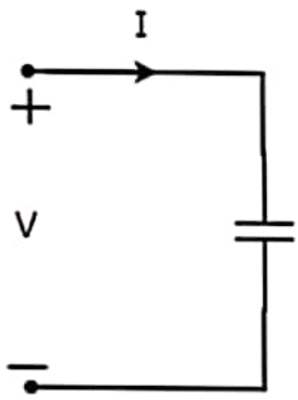
$$Q \alpha V$$

$$\Rightarrow Q = CV$$

Where,

- Q is the charge stored in the capacitor.
- C is the capacitance of a capacitor.

Let the current flowing through the capacitor is I amperes and the voltage across it is V volts. The symbol of capacitor along with current I and voltage V are shown in the following figure.



We know that the **current** is nothing but the **time rate of flow of charge**. Mathematically, it can be represented as

$$I = \frac{dQ}{dt}$$

Substitute $Q = CV$ in the above equation.

$$I = \frac{d(CV)}{dt}$$

$$\Rightarrow I = C\frac{dV}{dt}$$

$$\Rightarrow V = \frac{1}{C}\int Idt$$

From the above equations, we can conclude that there exists a linear relationship between voltage across capacitor and current flowing through it.

We know that power in an electric circuit element can be represented as

$$P = VI$$

Substitute $I = C\frac{dV}{dt}$ in the above equation.

$$P = V(C\frac{dV}{dt})$$

$$\Rightarrow P = CV\frac{dV}{dt}$$

By integrating the above equation, we will get the **energy** stored in the capacitor as

$$W = \frac{1}{2}CV^2$$

UNIT-1 (DC Circuits)

LECTURE NO:-1

Electrical Circuit Elements (R, L and C), Voltage and Current Sources

Resistance (R):

It may be defined as the property of a substance due to which it opposes (or or restricts) the electricity (i.e., electrons) through it.

Ohm 's Law This law applies to electric to electric conduction through good guides and may be stated as follows

The proportion of potential difference (V) between any two points on a guard to the current (I) flowing between them, is constant, fed the temperature of the guard does not change.

In other words, V/ I = constant or V/ I = R

where R is the resistance of the pilot between the two points considered.

Put in another way, it simply means that supplied R is kept constant, current is directly

proportionate to the possible difference across the ends of a pilot.

The practical unit of resistance is ohm. A pilot is said to have a resistance of one ohm

if it permits one ampere current to flow through it when one volt is impressed across its

outstations.

Inductance (L):

Inductance is the property of a material by virtue of which it opposes any change of magnitude and direction of electric current passing through operator. A rope of certain length, when twisted into a coil becomes a rudimental operator. A change in the magnitude of the current changes the electromagnetic field.

Increase in current expands the field& loss in current reduces it. A change in current produces change in the electromagnetic field. This induces a voltage across the coil according to Faradays laws of Electromagnetic Induction.

Induced Voltage V = L (di di/ dt)

Notes By: - Parshant Tyagi (EE Department)

V = Voltage across inductor in volts

I = Current through inductor in amps

Capacitance (C):

The property of a capacitor to 'store charge' may be called its capacitance.

As we may measure the capacity of a tank, not by the total mass or volume of water it can hold, but by the mass in kg of water took to raise its situation by one measure, likewise, the capacitance of a capacitor is defined as " the volume of charge wanted to result a unitp.d. between its plates. "

Suppose we give Q coulomb of charge to one of the two plate of capacitor and if ap.d. of V volts is established between the two, either its capacitance is

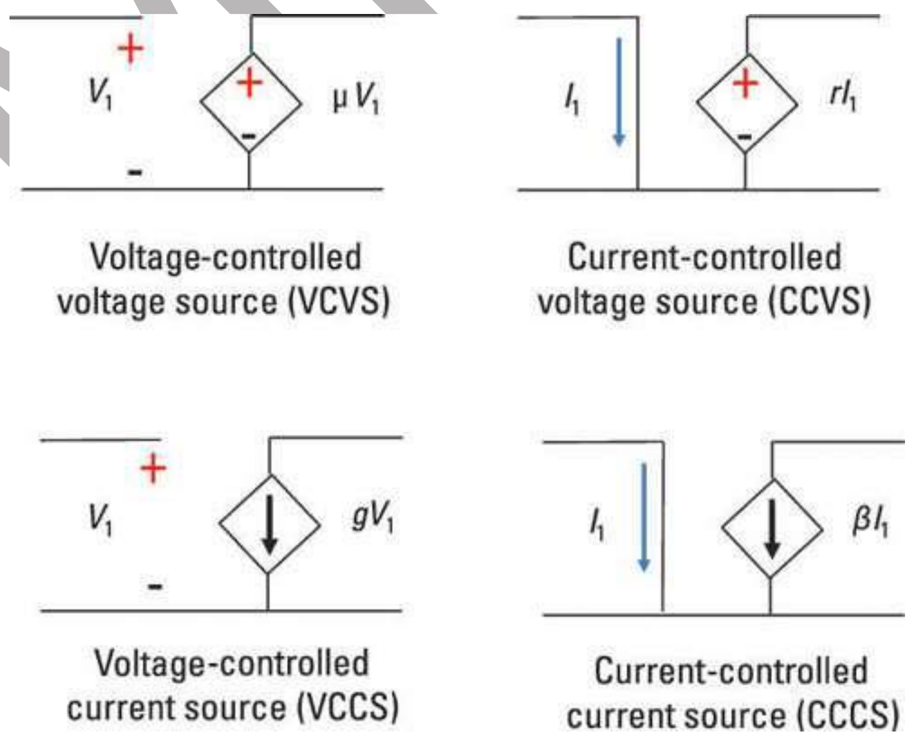
C = charge/ eventuality difference

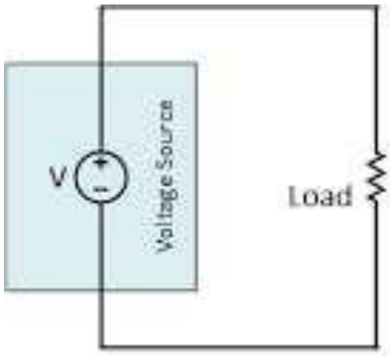
$$C = Q/ V$$

Hence, capacitance is the charge claimed per unit implicit difference.

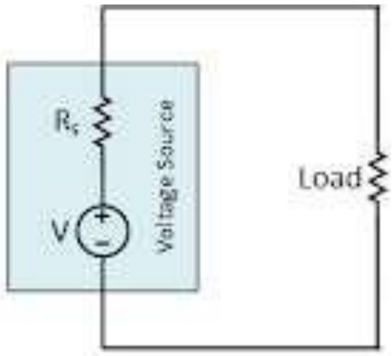
By description, the unit of capacitance is coulomb/ volt which is also called farad (in in honour of Michael Faraday)

Voltage and Current Sources:

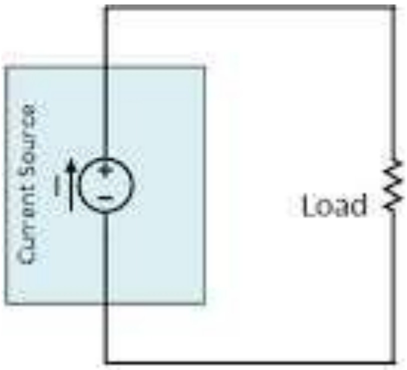




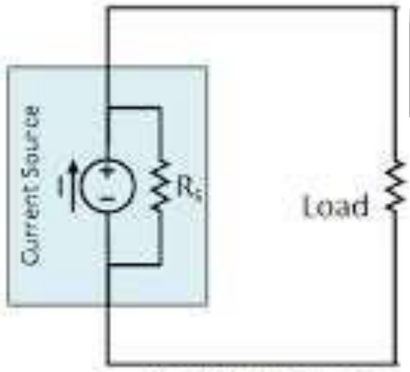
Ideal Voltage Source



Practical Voltage Source



Ideal Current Source



Practical Current Source

LECTURE NO:- 2

Kirchhoff Current and Voltage Laws, Series-Parallel Circuits

Kirchhoff s Current Law (KCL):

In any electrical network, the algebraic sum of the currents meeting at a point (or junction) is Zero.

That's the total current entering a junction is equal to the total current leaving that junction.

Consider the case of a network shown in Fig.

$$I1+(-I2)+(I3)+(+I4)+(-I5) = 0$$

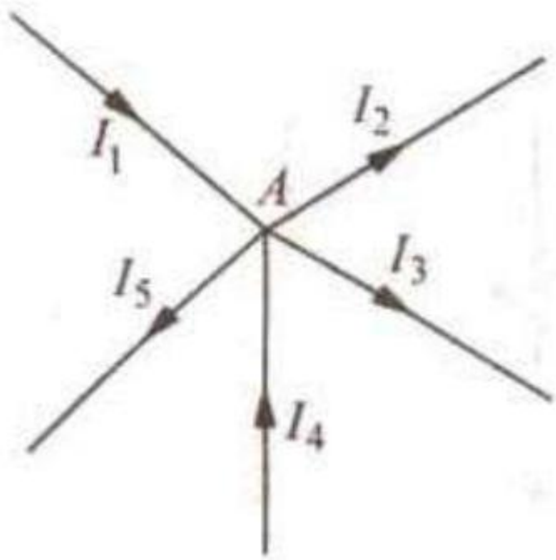
$$I1+I4-I2-I3-I5 = 0$$

Or

$$I1+I4 = I2+I3+I5$$

Or

Incoming currents =Outgoing currents



Kirchhoff's Voltage Law (KVL):

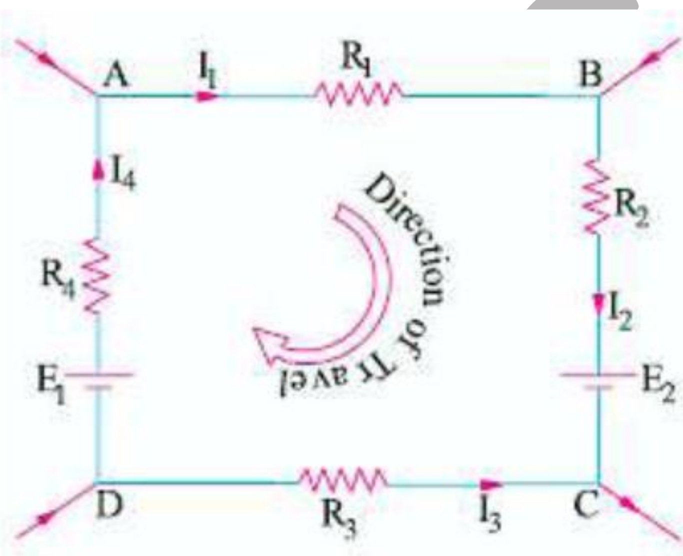
In any electrical network, the algebraic sum of the products of currents and resistances in each of the guides in any limited path (or mesh) in a network plus the algebraic sum of thee.m.f. ’s. in that path is zero.

That is, ΣIR + Σe.m.f = 0 round a mesh

It should be noted that algebraic sum is the sum which takes into account the contrariety of the voltage drops.

That is, if we start from a particular junction and go round the mesh till we come back to the starting point, either we must be at the same possibility with which we started.

Consider the free-for-all path ABCDA in Fig.

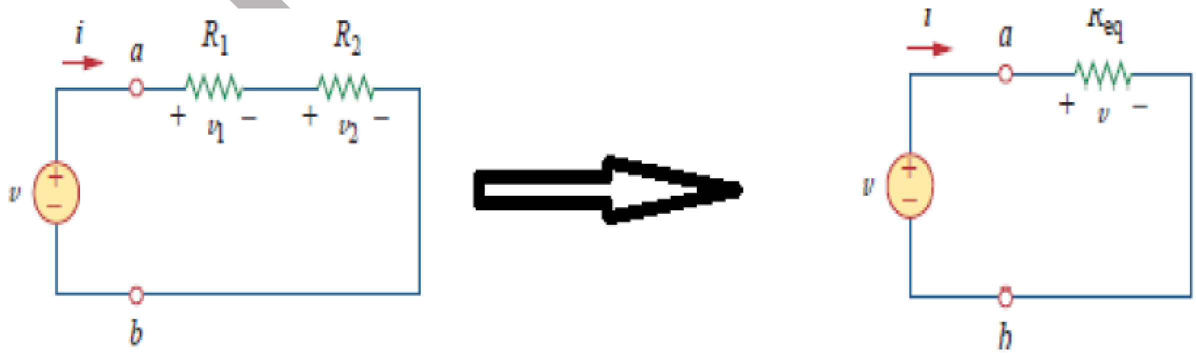


Using Kirchhoff's voltage law, we get

$$+I1R1 + I2R2 – I3R3 + I4\ R4 + E2 – E1 = 0$$

Or $$I1R1 + I2R2 – I3R3 + I4R4 = E1 –E2$$

Series Connection of Resistors:



Two or more resistors in a circuit are said to be in series when the current flowing through all the resistors is the same.

$v1 = iR1$....... (1) $v2 = iR2$........ (2)

If we apply KVL to the loop fig (b), we have

$-v + v1 + v2 = 0$.......................... (3)

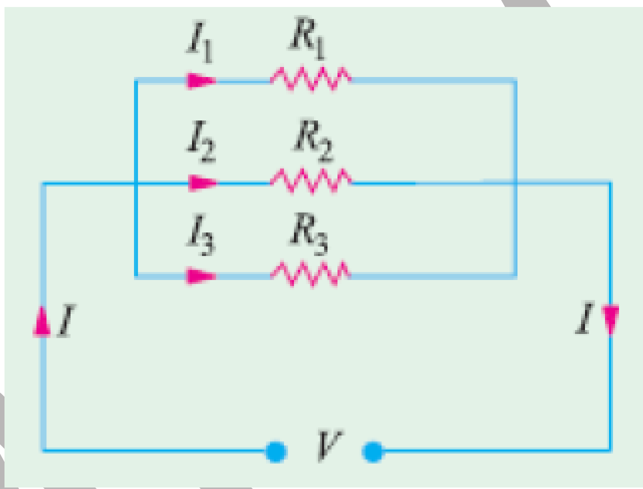
$v = v1 + v2 = I\,(R1 + R2)$

$i = v/\,(R1 + R2)$ ....(4)

$v = i\,Req$................ (5)

$Req = R1 + R2$

Parallel Connection of Resistors:



$$I = I_1 + I_2 + I_3 = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

Now, $I = \frac{V}{R}$ where V is the applied voltage.

R = equivalent resistance of the parallel combination.

$$\therefore \quad \frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3} \quad \text{or} \quad \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Also $G = G_1 + G_2 + G_3$

LECTURE NO:-3

Mesh Analysis:

A Mesh may be a loop which doesn't contain the other loops within it. Mesh Analysis provides another general procedure for analyzing circuits, using mesh currents as the circuit variables. Using mesh currents rather than element currents as circuit variables is convenient and\reduces the amount of equations that has got to be solved simultaneously. Recall that a loop may be a closed path with no node passed quite once. A mesh may be a loop that doesn't contain the other loop within it. Nodal analysis applies KCL to seek out unknown voltages during a given circuit, while mesh analysis applies KVL to seek out unknown currents. Mesh analysis isn't quite as general as nodal analysis because it's only applicable to a circuit that's planar. A planar circuit is one which will be drawn during a plane with no branches crossing one another; otherwise it's non-planar. A circuit may have crossing branches and still be planar if it are often redrawn such it's no crossing branches.

For example, the circuit in Fig.1(a) has two crossing branches, but it can be redrawn as in Fig. 2 Hence, the circuit in Fig.1 is planar.

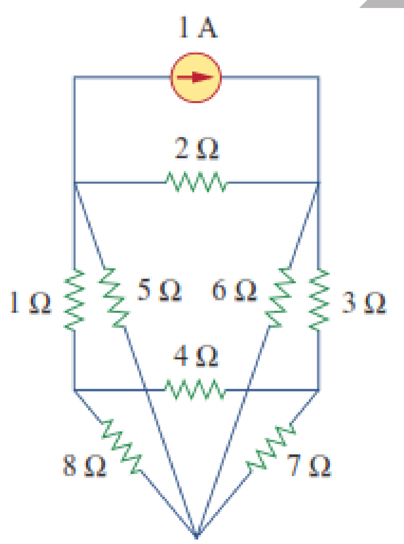


Fig1. Planner circuit with crossing branch

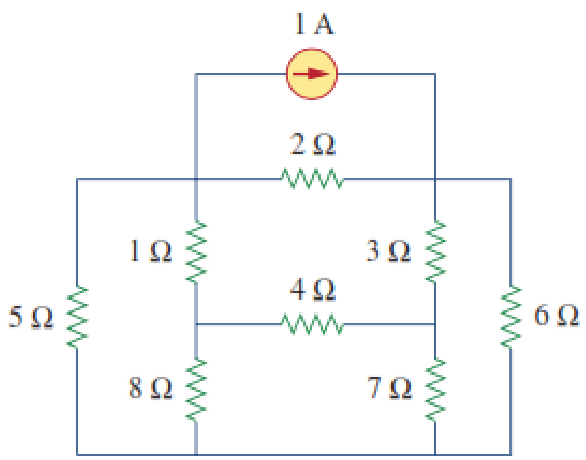


Fig2. Same circuit redrawn with no crossing

A mesh is a loop which does not contain any other loops within it. To understand mesh analysis, we should always first explain more about what we mean by a mesh.

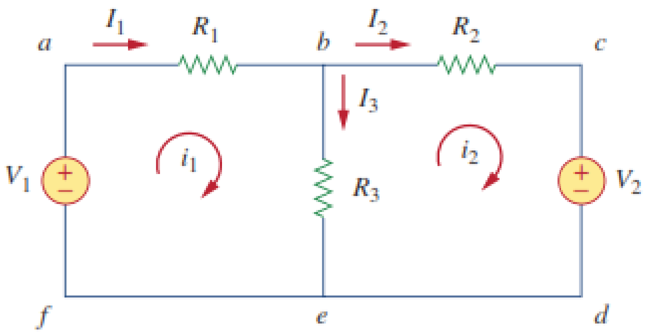


Fig.3 A Circuit with 2 Meshes

In Fig.3, for instance , paths abefa and bcdeb are meshes, but path abcdefa isn't a mesh. The current through a mesh is understood as mesh current. In mesh analysis, we have an interest during a pplying KVL to seek out the mesh currents in a given circuit.

In this section, we'll apply mesh analysis to planar circuits that don't contain current sources. In the next section, we'll consider circuits with current sources. In the mesh analysis of a circuit with n meshes, we take the subsequent three steps.

Steps to Determine Mesh Currents:

1. Assign mesh currents $i_1, i_2, \ldots, i_n$ to the n meshes.
2. Apply KVL to each of the n meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
3. Solve the resulting n simultaneous equations to get the mesh currents.

To illustrate the steps, consider the circuit in Fig.3. The first step requires that mesh currents and are assigned to meshes 1 and 2. Although a mesh current could also be assigned to every mesh in an arbitrary direction, it's conventional to assume that every mesh current flows clockwise. As the second step, we apply KVL to every mesh. Applying KVL to mesh 1, we obtain

$$-V_1 + R_1 i_1 + R_3(i_1 - i_2) = 0$$

or

$$(R_1 + R_3)i_1 - R_3 i_2 = V_1 \qquad \text{Equation.1}$$

For Mesh 2, Applying KVL gives,

$$R_2 i_2 + V_2 + R_3(i_2 - i_1) = 0$$

or

$$-R_3 i_1 + (R_2 + R_3) i_2 = -V_2$$ Equation.2

Note in Eq.1 that the coefficient of I1 the sum of the resistances in the first mesh, while the coefficient of I2 is the negative of the resistance common to meshes 1 and 2. Now observe that the same is true in Eq.2. This can function a shortcut way of writing the mesh equations. We will exploit this concept in Section.

The third step is to unravel for the mesh currents. Putting Eqs. 1 and a couple of in matrix form yields

$$\begin{bmatrix} R_1 + R_3 & -R_3 \\ -R_3 & R_2 + R_3 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ -V_2 \end{bmatrix}$$

which can be solved to get the mesh currents and that we are at liberty to use any technique for solving the equation . According to Eq. if a circuit has n nodes, b branches, and l independent loops or meshes, then Hence, independent equation are required to unravel the circuit using mesh analysis.

Problem1. For the circuit in Fig.1, find the branch currents and using mesh analysis.

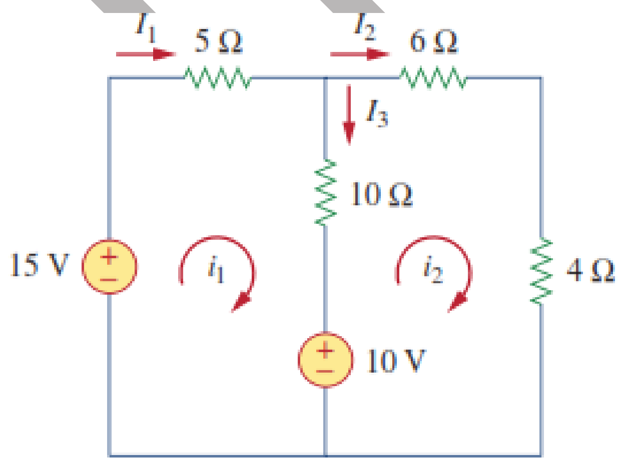


Fig.1

We first obtain the mesh currents using KVL. For mesh 1,

$$-15 + 5i_1 + 10(i_1 - i_2) + 10 = 0$$

or

$$3i_1 - 2i_2 = 1$$ Eq.1

For Mesh 2,

$$6i_2 + 4i_2 + 10(i_2 - i_1) - 10 = 0$$

or

$$i_1 = 2i_2 - 1$$

Eq.2

By using substitution Method, we substitute equation 2 into equation 1, we get

$$6i_2 - 3 - 2i_2 = 1 \quad \Rightarrow \quad i_2 = 1 \text{ A}$$

From eq.2

$$i_1 = 2i_2 - 1 = 2 - 1 = 1 \text{ A. Thus,}$$

$$I_1 = i_1 = 1 \text{ A}, \quad I_2 = i_2 = 1 \text{ A}, \quad I_3 = i_1 - i_2 = 0$$

LECTURE NO:-4

Node Analysis

NODE ANALYSIS:

Nodal analysis provides a general procedure for analyzing circuits using node voltages because the circuit variables. Choosing node voltages rather than element voltages as circuit variables is convenient and reduces the amount of equations one must solve simultaneously. To simplify matters, we shall assume during this section that circuits don't contain voltage sources. Circuits that contain voltage sources are going to be analyzed within the next section.

In nodal analysis, we have an interest find the node voltages. Given a circuit with n nodes without voltage sources, the nodal analysis of the circuit involves taking the subsequent three steps.

Steps to Determine Node Voltages:

1. Select a node as the reference node. Assign voltages v_1, $v_2, \ldots, v_{n-1}$ to the remaining $n - 1$ nodes. The voltages are referenced with respect to the reference node.
2. Apply KCL to each of the $n - 1$ nonreference nodes. Use Ohm's law to express the branch currents in terms of node voltages.
3. Solve the resulting simultaneous equations to obtain the unknown node voltages.

We shall now explain and apply these three steps.

1. The first step in nodal analysis is selecting a node as the reference or datum node. The reference node is usually called the bottom

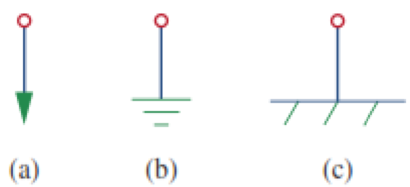


Fig1. Common symbols for indicating reference node, (a) common ground, (b) ground, (c) chassis ground.

or chassis acts as a point of reference for all circuits. Once we've selected a reference node, we assign voltage designations to non reference nodes. Consider, for instance , the circuit in Fig.2(a). Node 0 is that the reference node while nodes 1 and a couple of are assigned voltages and respectively. Keep in mind that the node voltages are defined with reference to the reference node. As illustrated in Fig.1(a), each node voltage is the voltage rise from the reference node to the corresponding non reference node or simply the voltage of that node with respect to the reference node. As the second step, we apply KCL to every non reference node within the circuit. To avoid putting an excessive amount of information on an equivalent circuit, the circuit in Fig.2(a) is redrawn in Fig.2(b), where we now add and because the currents through resistors and respectively.

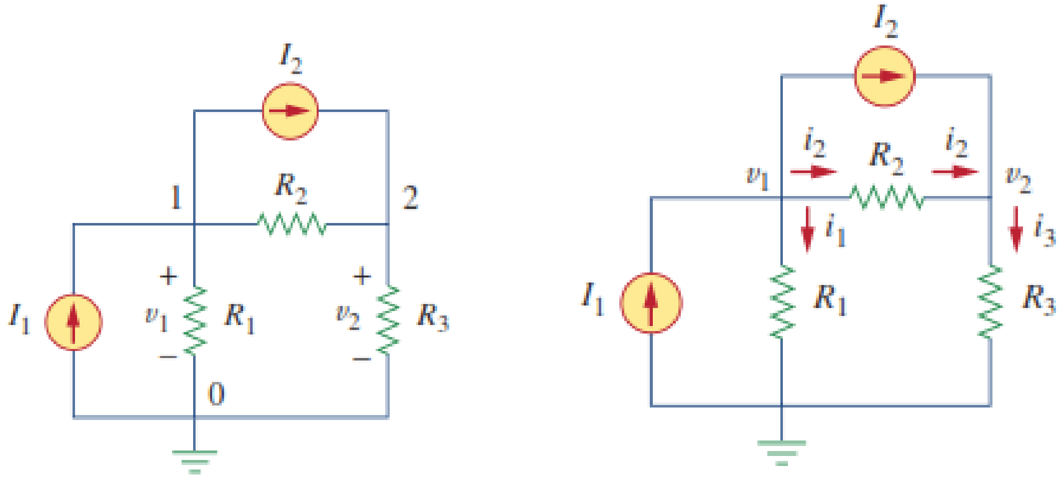


Fig2. Typical Circuit for Nodal Analysis

At Node 1, applying KCL Gives

$$I_1 = I_2 + i_1 + i_2$$

At Node 2,

$$I_2 + i_2 = i_3$$

We now apply Ohm's law to precise the unknown currents and in terms of node voltages. The key idea in touch in mind is that, since resistance may be a passive element, by the passive sign convention, current should be due a better potential to a lower potential. Current flows from a better potential to a lower potential during a resistor.

We can express this Principle as

$$\boxed{i = \frac{v_{\text{higher}} - v_{\text{lower}}}{R}}$$

Note that this principle is in agreement with the way we defined resistance. With this in mind, we obtain from Fig.2(b),

$$i_1 = \frac{v_1 - 0}{R_1} \quad \text{or} \quad i_1 = G_1 v_1$$

$$i_2 = \frac{v_1 - v_2}{R_2} \quad \text{or} \quad i_2 = G_2(v_1 - v_2)$$

$$i_3 = \frac{v_2 - 0}{R_3} \quad \text{or} \quad i_3 = G_3 v_2$$

Substituting above equation in Equation .of KCL results, respectively, in

$$I_1 = I_2 + \frac{v_1}{R_1} + \frac{v_1 - v_2}{R_2}$$

$$I_2 + \frac{v_1 - v_2}{R_2} = \frac{v_2}{R_3}$$

Problem1. Calculate the node voltages in the circuit shown in Fig. 1(a).

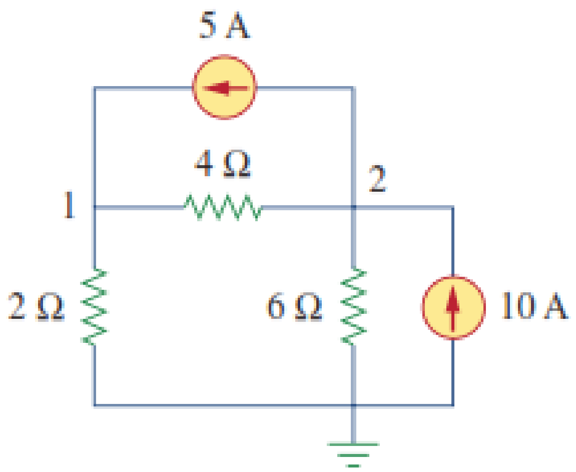


Fig. 1(a)

Consider Fig.1(b), where the circuit in Fig. 1(a) has been prepared for nodal analysis. Notice how the currents are selected for the appliance of KCL. Except for the branches with current sources, the labeling of the currents is bigoted but consistent. (By consistent, we mean that if, for example, we assume that enters the resistor from the left-hand side, must leave the resistor from the right-hand side.) The reference node is selected, and therefore the node voltages and are now to be determined.

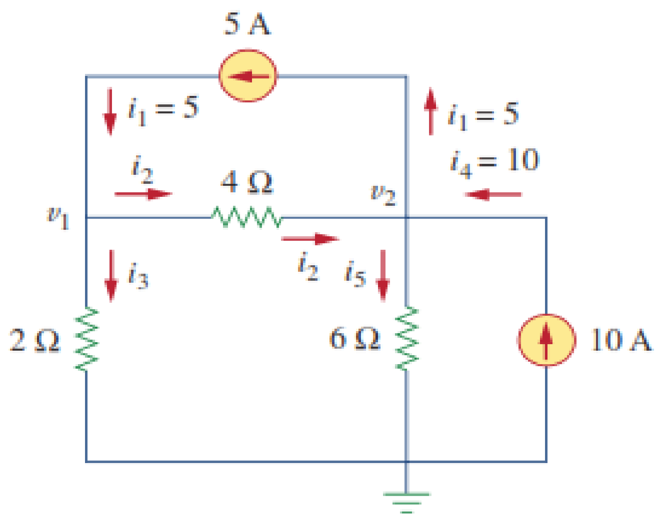


Fig.1(b)

At node 1, applying KCL and Ohm's law gives

$$i_1 = i_2 + i_3 \quad \Rightarrow \quad 5 = \frac{v_1 - v_2}{4} + \frac{v_1 - 0}{2}$$

Multiplying each term within the last equation by 4, we obtain

$$20 = v_1 - v_2 + 2v_1$$

or

$$3v_1 - v_2 = 20 \qquad \text{Eq.1}$$

At node 2, we do an equivalent thing and obtain

$$i_2 + i_4 = i_1 + i_5 \quad \Rightarrow \quad \frac{v_1 - v_2}{4} + 10 = 5 + \frac{v_2 - 0}{6}$$

Multiplying each term by 12 results in

$$3v_1 - 3v_2 + 120 = 60 + 2v_2$$

or

$$-3v_1 + 5v_2 = 60 \qquad \text{Eq.2}$$

Now for solving equation 1 & 2, we get value of V1 and V2

V1= 13.33 V, V2= 20 V Ans.

LECTURE NO:-5

Thevenin's Theorem

Introduction:

Any complicated network i.e. several sources, multiple resistors are present if the only element response is desired then use the network theorems. Network theorems also are are often termed as network reduction techniques. Each and each theorem got its importance of solving network. allow us to see some important theorems with DC and AC excitation with detailed procedures.

Thevenin's Theorem:

Thevenin's Theorem and Norton's theorem are two important theorems in solving Network problems having many active and passive elements. Using these theorems the networks are often reduced to simple equivalent circuits with one active source and one element. In circuit analysis many an times the present through a branch is required to be found when it's value is modified with all other element values remaining same. In such cases checking out whenever the branch current using the traditional mesh and node analysis methods is sort of awkward and time consuming. But with the straightforward equivalent circuits (with one active source and one element) obtained using these two theorems the calculations become very simple. Thevenin's and Norton's theorems are dual theorems.

Thevenin's Theorem Statement :

Any linear, bilateral two terminal network consisting of sources and resistors(Impedance),can be replaced by the same circuit consisting of a voltage source serial with a resistance (Impedance).The equivalent voltage source VTh is that the circuit voltage looking into the terminals(with concerned branch element removed) and therefore the equivalent resistance RTh while all sources are replaced by their internal resistors at ideal condition i.e. voltage source is brief circuit and current source is circuit .

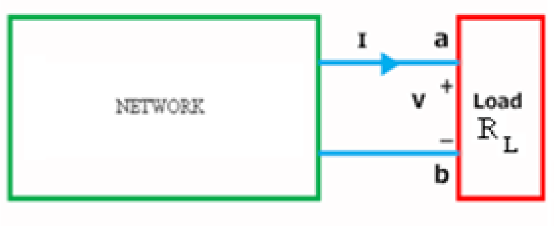


Fig. (a)

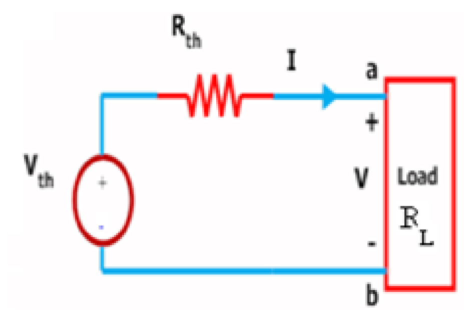


Fig.(b)

Figure (a) shows an easy block representation of a network with several active / passive elements with the load resistance RL connected across the terminals 'a & b' and figure (b) shows the Thevenin equivalent circuit with VTh connected across RTh& RL . Main steps to seek out outVTh and RTh :

1. The terminals of the branch/element through which the present is to be acknowledged are marked as say a & b after removing the concerned branch/element.

2. circuit voltage VOC across these two terminals is acknowledged using the traditional network mesh/node analysis methods and this is able to be VTh .

3. Thevenin resistance RTh is acknowledged by the tactic depending upon whether the network contains dependent sources or not. a. With dependent sources: RTh = Voc / Isc b. Without dependent sources : RTh = Equivalent resistance looking into the concerned terminals with all voltage & current sources replaced by their internal impedances (i.e. ideal voltage sources short circuited and ideal current sources open circuited)

4. Replace the network with VTh serial with RTh and therefore the concerned branch resistance (or) load resistance across the load terminals(A&B) as shown in below fig.

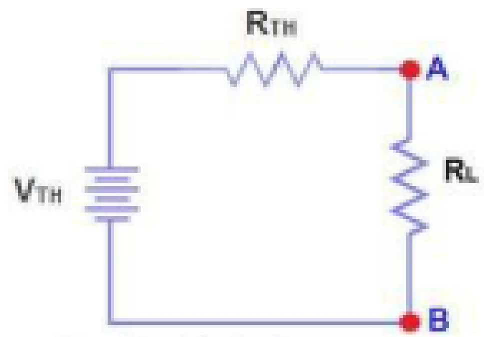
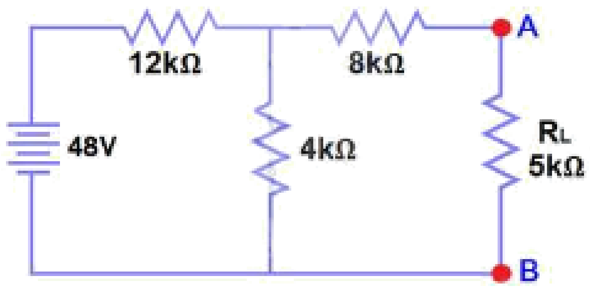


Fig.Thevenins Theorem Equivalent circuit

Example: Find VTH, RTH and therefore the load current and cargo voltage flowing through RL resistor as shown in fig. by using Thevenin's Theorem?



Solution: The resistance RL is removed and therefore the terminals of the resistance RL are marked as A & B as shown within the fig. (1)

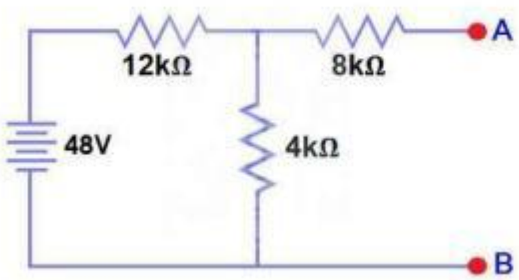


Fig.1

Calculate / measure the circuit Voltage. this is often the Thevenin Voltage (VTH). we've already removed the load resistor from fig.(a), therefore the circuit became an circuit as shown in fig (1). Now we've to calculate the Thevenin's Voltage. Since 3mA Current flows in both 12kΩ and 4kΩ resistors as this is often a circuit because current won't flow within the 8kΩ resistor because it is open. So 12V (3mA x 4kΩ) will appear across the 4kΩ resistor. We also know that current isn't flowing through the 8kΩ resistor because it is circuit , but the 8kΩ resistor is in parallel with 4k resistor. therefore the same voltage (i.e. 12V) will appear across the 8kΩ resistor as 4kΩ resistor. Therefore 12V will appear across the AB terminals. So, VTH = 12V

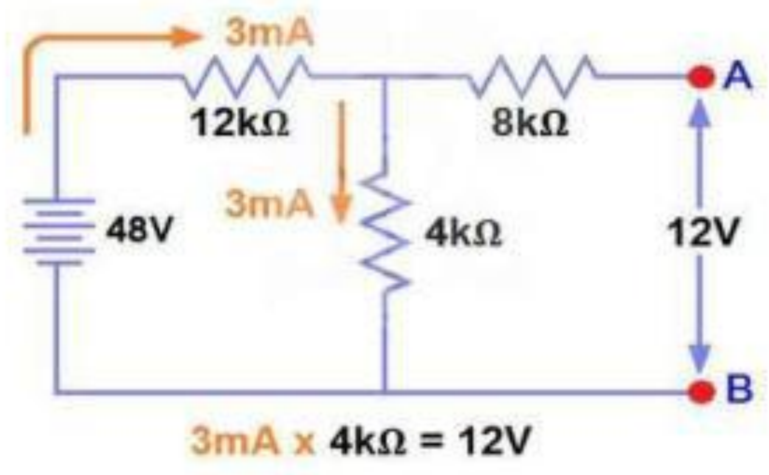


Fig.2

All voltage & current sources replaced by their internal impedances (i.e. ideal voltage sources short circuited and ideal current sources open circuited) as shown in fig.(3)

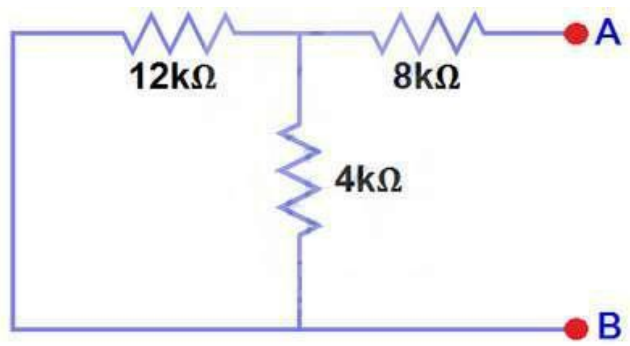
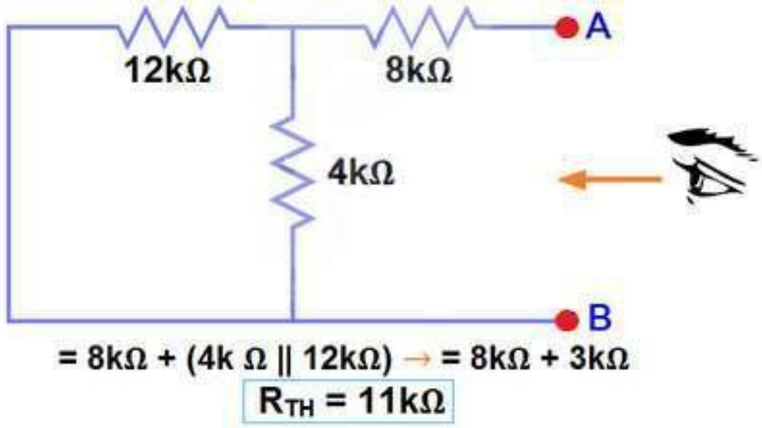
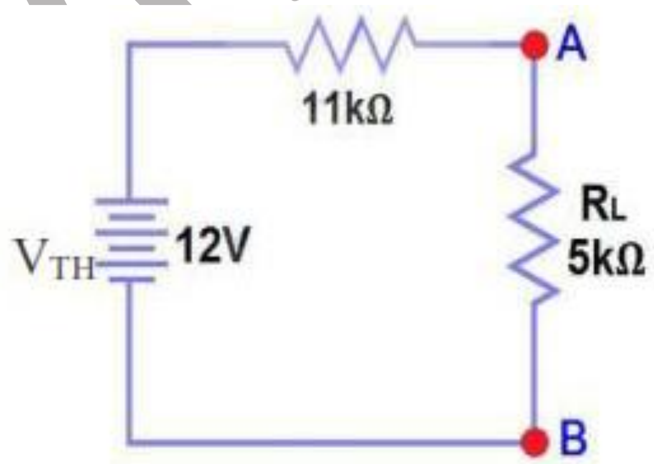


Fig.3

Calculate /measure the circuit Resistance. this is often the Thevenin Resistance (RTH)We have Reduced the 48V DC source to zero is like replace it with a brief circuit as shown in figure (3) we will see that 8kΩ resistor is serial with a parallel connection of 4kΩ resistor and 12k Ω resistor. i.e.:





Now apply Ohm's law and calculate the total load current from fig 5. IL = VTH/ (RTH + RL)= 12V / (11kΩ + 5kΩ) = 12/16kΩ

IL= 0.75mA And VL = ILx RL= 0.75mA x 5kΩ **VL= 3.75V Answer**

LECTURE NO:-6

Norton's Theorem

Norton's Theorem Statement:

Any linear, bilateral two terminal network consisting of sources and Resistors(Impedance),can be replaced by the same circuit consisting of a current source in parallel with a resistance (Impedance),the current source being the short circuited current across the load terminals and therefore the resistance being the interior resistance of the source network rummaging through the open circuited load terminals.

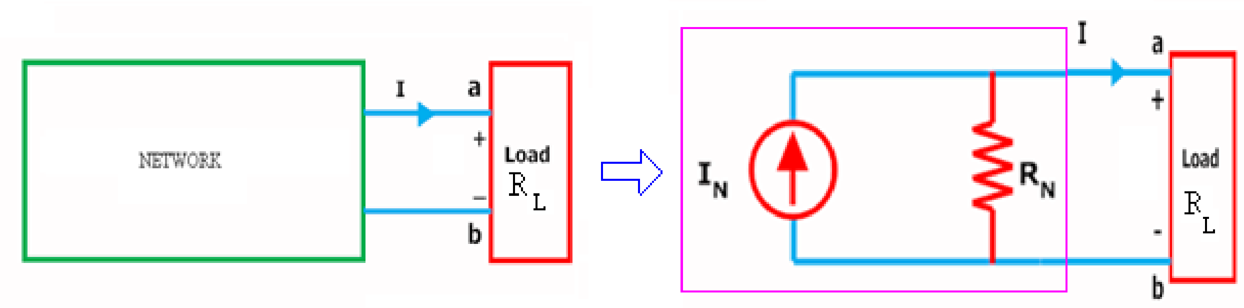


Fig.(a) Fig.(b)

Figure (a) shows an easy block representation of a network with several active / passive elements with the load resistance RL connected across the terminals 'a & b' and figure (b) shows the Norton equivalent circuit with IN connected across RN &RL .

Main steps to seek out call at and RN:

1. The terminals of the branch/element through which the present is to be acknowledged are marked as say a & b after removing the concerned branch/element.
2. circuit voltage VOC across these two terminals and ISC through these two terminals are acknowledged using the traditional network mesh/node analysis methods and that they are same as what we obtained in Thevenin's equivalent circuit.

3. Next Norton resistance RN is acknowledged depending upon whether the network contains dependent sources or not. a) With dependent sources: RN = Voc / Isc

b) Without dependent sources : RN = Equivalent resistance looking into the concerned terminals with all voltage & current sources replaced by their internal impedances (i.e. ideal voltage sources short circuited and ideal current sources open circuited)

4. Replace the network with IN in parallel with RN and therefore the concerned branch resistance across the load terminals(A&B) as shown in below fig

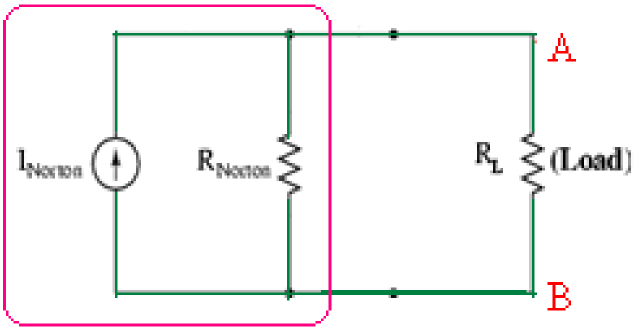


Fig.Nortan Equivalent Circuit

Example: Find the present through the resistance RL (1.5 Ω) of the circuit shown within the figure (a) below using Norton's equivalent circuit?

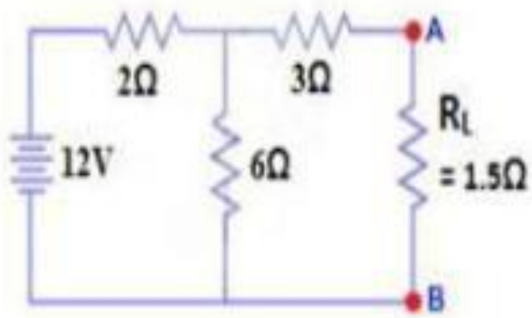


Fig.(a)

Solution: To seek out out the Norton's equivalent ckt we've to seek out call at = Isc ,RN=Voc/ Isc. Short the 1.5Ω load resistor as shown in (Fig 1), and Calculate / measure the short Current. this is often the Norton Current (IN).

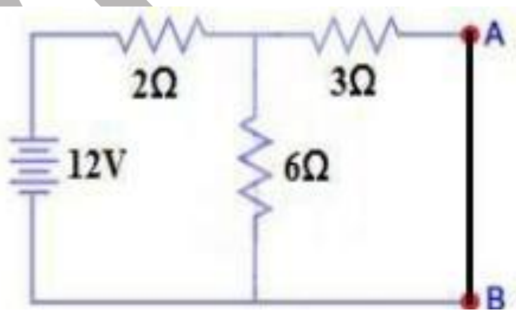


Fig.1

we've shorted the AB terminals to work out the Norton current, IN. The 6Ω and 3Ω are then in parallel and this parallel combination of 6Ω and 3Ω are then serial with 2Ω.

So the Total Resistance of the circuit to the Source is:- 2Ω + (6Ω || 3Ω) ….. (|| = in parallel with)

$$RT = 2\Omega + [(3\Omega \times 6\Omega) / (3\Omega + 6\Omega)]$$

$$RT = 2\Omega + 2\Omega \quad RT = 4\Omega$$

$$IT = V / RT$$

$$IT = 12V / 4\Omega = 3A..$$

Now we've to seek out ISC = IN… Apply CDR… (Current Divider Rule)…

$$ISC = IN = 3A \times [(6\Omega / (3\Omega + 6\Omega)] = 2A.$$

$$ISC = IN = 2A.$$

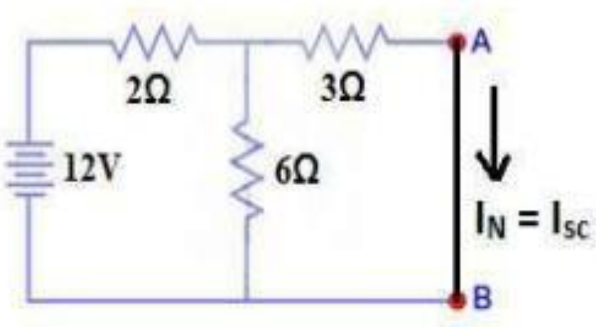


Fig.2

All voltage & current sources replaced by their internal impedances (i.e. ideal voltage sources short circuited and ideal current sources open circuited) and Open Load Resistor. as shown in fig.(4)

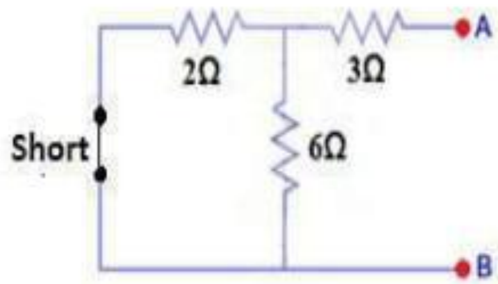


Fig.3

Calculate measure the circuit Resistance, this is often called Norton Resistance. we've to scale back the 12 V Source to zero is like is like replace with short as shown within the figure(4).

$3\Omega + (6\Omega \parallel 2\Omega)$ ….. ($\parallel$ = in parallel with)

$$R_N = 3\Omega + [(6\Omega \times 2\Omega) / (6\Omega + 2\Omega)]$$

$$R_N = 3\Omega + 1.5\Omega$$

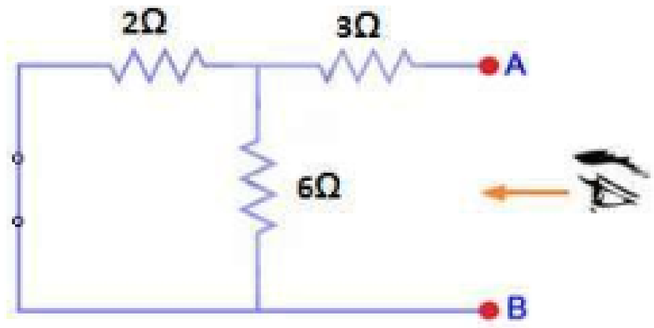


Fig.4

Connect the RN in Parallel with Current Source IN and re-connect the load resistor. this is often shown in fig (5) i.e. Norton Equivalent circuit with load resistor.

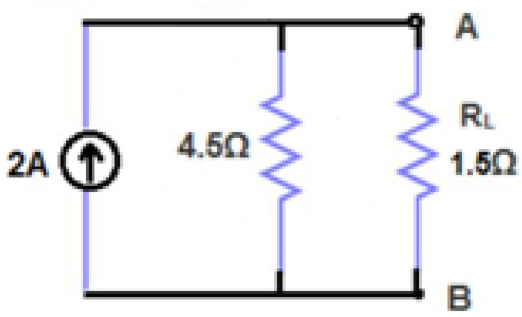


Fig.5

Now apply the Ohm's Law and calculate the load current through Load resistance across the terminals A&B.

Load Current through Load Resistor is IL = IN x [RN / (RN+ RL)]

IL= 2A x (4.5Ω /4.5Ω +1.5kΩ)

IL = 1. 5A Ans.

LECTURE NO:- 7

Maximum Power Transfer Theorem

MAXIMUM POWER TRANSFER THEOREM:

When designing a circuit, it's often important to be ready to answer one among the subsequent questions:

What load should be applied to a system to make sure that the load is receiving maximum power from the system?

Conversely:

For a specific load, what conditions should be imposed on the source to make sure that it'll deliver the utmost power available?

Even if a load can't be set at the worth that might end in maxi- mum power transfer, it's often helpful to possess some idea of the worth which will draw maximum power in order that you'll compare it to the load at hand. as an example , if a design involves a load of 100 Ω, to make sure that the load receives maximum power, employing a resistor of 1 Ω or 1 kΩ.

Results in an influence transfer that's much but the utmost possible. However, employing a load of 82 Ω or 120 Ω probably leads to a reasonably good level of power transfer. Fortunately, the method of finding the load which will receive maximum power from a specific system is sort of straightforward thanks to the

Maximum power transfer theorem, which states the following:

A load will receive maximum power from a network when its resistance is strictly adequate to the Thévenin resistance of the network applied to the load;

$$\boxed{R_L = R_{Th}}$$

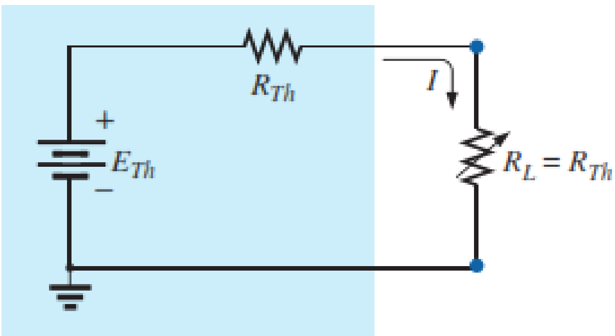


Fig.1

In other words, for the Thévenin equivalent circuit in Fig.1 when the load is about adequate to the Thévenin resistance, the load will receive maximum power from the network

Using Fig 1, with RL=RTH, we will determine the utmost power delivered to the load by first finding the current:

$$I_L = \frac{E_{Th}}{R_{Th} + R_L} = \frac{E_{Th}}{R_{Th} + R_{Th}} = \frac{E_{Th}}{2R_{Th}}$$

Then we substitute into the facility equation:

$$P_L = I_L^2 R_L = \left(\frac{E_{Th}}{2R_{Th}}\right)^2 (R_{Th}) = \frac{E_{Th}^2 R_{Th}}{4R_{Th}^2}$$

and

$$\boxed{P_{L_{max}} = \frac{E_{Th}^2}{4R_{Th}}}$$

To demonstrate that maximum power is indeed transferred to the load under the conditions defined above, consider the Thévenin equivalent circuit in Fig.2.Before stepping into detail, however, if you were to guess what value of RL would end in maximum power transfer to RL you would possibly think that the smaller the worth of RL

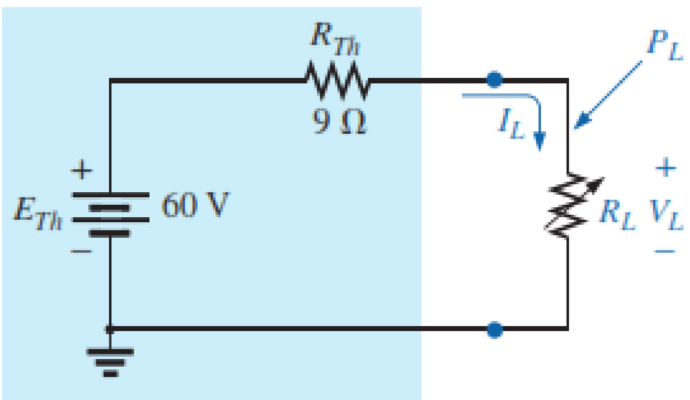


Fig.2- Thevenin's equivalent network

The better it's because the present reaches a maximum when it's squared within the power equation. the matter is, however, that within the equation P=I2RL

The load resistance may be a multiplier. because it gets smaller, it forms a smaller product. but , you would possibly suggest larger values of because the output voltage increases, and power is decided by P=V2/RL resistance is within the denominator of the equation and causes the resulting power to decrease. A balance must obviously be made between the load resistance and therefore the resulting current or voltage. the subsequent discussion shows that:

Maximum power transfer occurs when the load voltage and current are one-half their maximum possible values.

For the circuit in Fig.2 , the current through the load is

$$I_L = \frac{E_{Th}}{R_{Th} + R_L} = \frac{60\ \text{V}}{9\ \Omega + R_L}$$

The voltage is calculated by

$$V_L = \frac{R_L E_{Th}}{R_L + R_{Th}} = \frac{R_L(60\ \text{V})}{R_L + R_{Th}}$$

And the Power is

$$P_L = I_L^2 R_L = \left(\frac{60\ \text{V}}{9\ \Omega + R_L}\right)^2 (R_L) = \frac{3600 R_L}{(9\ \Omega + R_L)^2}$$

LECTURE NO:-8

Superposition Theorem

Superposition Theorem:

The principle of superposition helps us to research a linear circuit with quite one current or voltage sources sometimes it's easier to seek out out the voltage across or current during a branch of the circuit by considering the effect of 1 source at a time by replacing the opposite sources with their ideal internal resistances.

Superposition Theorem Statement:

Any linear, bilateral two terminal network consisting of quite one sources, the entire current or voltage in any a part of a network is adequate to the algebraic sum of the currents or voltages within the required branch with each source acting individually while other sources are replaced by their ideal internal resistances. (i.e. Voltage sources by a brief circuit and current sources by open circuit)

Steps to use Super position Principle:

1. Replace all independent sources with their internal resistances except one source. Find the output (voltage or current) thanks to that active source using nodal or mesh analysis.

2. Repeat step 1 for every of the opposite independent sources.

3. Find the entire contribution by adding algebraically all the contributions thanks to the independent sources.

Example: By Using the superposition theorem find I within the circuit shown in figure?

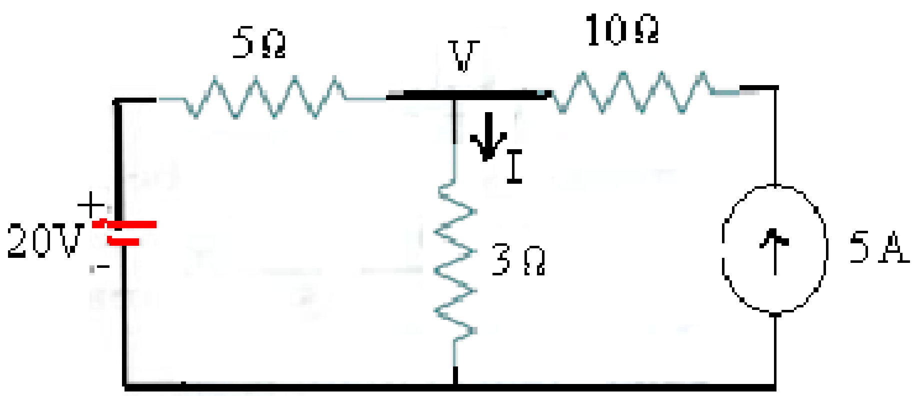


Fig.a

Solution:

Applying the superposition theorem, the present I2 within the resistance of three Ω to the voltage source of 20V alone, with current source of 5A open circuited.

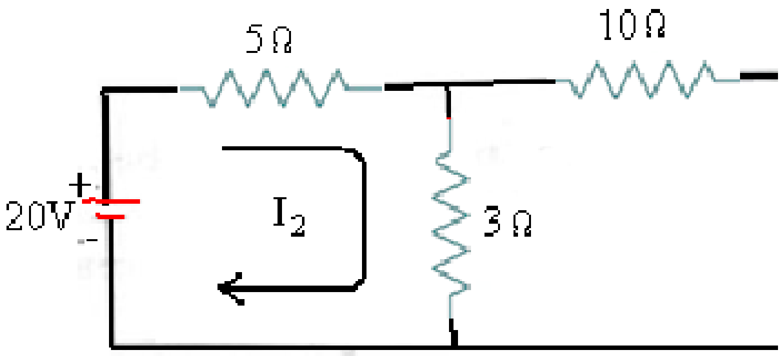


Fig.1

I2 = 20/(5+3) = 2.5A

Similarly the present I5 within the resistance of three Ω thanks to the present source of 5A alone with voltage source of 20V short circuited [as shown within the figure.2 below] is given by :

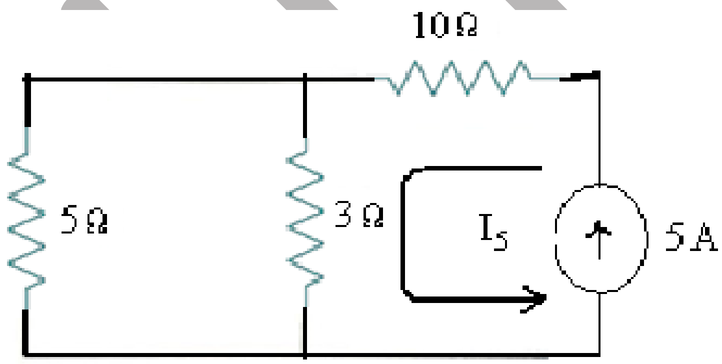


Fig.2

I5= 5 x 5/(3+5) = 3.125 A

Total current passing through the resistance of 3Ω is then = I2 + I5= 2.5 + 3.125 = **5.625 A**Let us to verify the answer using the essential nodal analysis pertaining to the node marked with V in fig.(a).Then we get :

$$\frac{V-20}{5}+\frac{V}{3}=5$$

3V-60+5V=15× 5

8V-60=75

8V=135

V=16.875

The current I passing through the resistance of 3Ω =V/3 = 16.875/3 = **5.625 A.**

References:

- https://wiraelectrical.com/mesh-currentanalysis/#:~:
- text=A%20mesh%20is%20a%20loop%20which%20does%20not%20contain%20any,(loop%201%20and%202).
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